

VOLUME 03

Enterprise Data Platform

Enterprise Google Cloud Data Platform Blueprint Series

Fortune 100 Reference Architecture and Implementation Baseline

Architecture Baseline: 1.0

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Classification: Enterprise Reference Architecture

Audience: Chief Data Officer, CIO, CTO, Enterprise Architecture Review Board, Data Governance Council, Security Architecture Review Board, Cloud Center of Excellence, platform engineering and domain data teams

Document Control

This volume defines the enterprise data platform that runs on the Google Cloud landing zone established by Volume 02. It is a reference architecture, not a substitute for workload discovery, regulatory assessment or service-specific validation. The design assumes a global Fortune 100 enterprise and establishes mandatory platform patterns, adaptable patterns, architecture decisions, implementation standards and acceptance evidence.

Field	Value
Document	Volume 03 - Enterprise Data Platform
Status	Architecture baseline for adaptation and implementation
Version	1.0
Date	11 July 2026
Owner	Enterprise Data Platform Architecture
Approvers	CDO, Enterprise Architecture, Security, Privacy, Platform Engineering, FinOps
Review cycle	Quarterly and on material Google Cloud product change
Related volumes	Volume 01 Executive Architecture Guide; Volume 02 Google Cloud Landing Zone; Volume 04 Data Engineering; Volume 05 Data Mesh; Volume 07 Security Blueprint; Volume 10 Operations SRE
Primary standards	Google Cloud Architecture Framework, Zero Trust, data mesh, lakehouse, SRE, FinOps, privacy by design

Current Product Terminology

Google Cloud product naming evolves. This volume uses the current product names available at publication and preserves former names in parentheses where they remain common in enterprise programs. **Knowledge Catalog** is the current name for the governance capability formerly called Dataplex Universal Catalog. **BigQuery sharing** is the current name for the capability formerly called Analytics Hub. **Apache Iceberg managed tables** is the current name for the BigQuery-managed Iceberg table capability formerly described as BigLake tables for Apache Iceberg in BigQuery. Teams must validate preview status, regional availability, quotas and feature compatibility during implementation.

How to Use This Volume

The volume is organized from capability and source architecture through ingestion, storage, governance, sharing, security, operations and production deployment. Each design statement is classified as one of four types:

Classification	Meaning
Mandatory baseline	Required for all production domains unless an approved exception exists.
Recommended pattern	The default for most workloads; another pattern requires documented engineering justification.
Conditional pattern	Use only when workload characteristics meet the stated selection criteria.
Future capability	Included in the target architecture but not required for the first platform release.

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Executive Summary

The enterprise data platform provides a governed path from hundreds of operational, SaaS, event, file, mainframe and cross-cloud sources to trusted data products used by analytics, operational applications and AI. The reference design supports 10-50 petabytes of historical data, 50-250 terabytes of new data per day, billions of events per day and globally distributed consumers without creating a single monolithic project or a central delivery bottleneck.

The target architecture combines Cloud Storage for immutable and open-format data, BigQuery for governed analytical products, BigLake and Apache Iceberg where open-engine interoperability is a real requirement, Pub/Sub and Dataflow for event processing, Datastream for supported database change data capture, Knowledge Catalog for enterprise metadata and governance, and BigQuery sharing for controlled publication. Bigtable, Spanner and AlloyDB are not alternate warehouses; they are conditional operational data stores selected for specific low-latency, globally consistent or PostgreSQL-compatible application patterns.

The operating model is federated. A central platform team owns reusable ingestion frameworks, shared governance, security controls, observability, cost attribution and paved-road automation. Domain teams own data products, contracts, quality, semantic meaning and consumer commitments. The architecture separates control planes from data planes and separates business ownership from infrastructure administration.

The design deliberately avoids treating every workload as streaming, every object as an Iceberg table or every dataset as a universal enterprise product. Latency, update pattern, interoperability, privacy, recovery and unit economics drive the decision. Batch is the default when it satisfies business latency. Streaming is reserved for time-sensitive decisions or event-driven integration. BigQuery native tables are the default analytical serving format. Apache Iceberg is selected when multiple engines must read and write a common table or customer-controlled object storage is a requirement.

Reference Enterprise Assumptions

Area	Reference assumption	Architecture implication
Scale	25 PB current estate; 120 TB daily ingestion; 5 billion events per day	Storage lifecycle and compute isolation are mandatory.
Sources	250 enterprise sources and more than 2,000 feeds	Source onboarding must be metadata-driven and automated.
Consumers	4,000 analysts, 600 data engineers/scientists and 300 application integrations	Access and workload isolation cannot rely on manual grants.
Geography	AMER, EMEA and APAC data-residency boundaries	Regional data planes with global governance metadata.
Availability	Tiered RTO/RPO and 99.9%-99.99% service objectives	Replay, backup, regional recovery and IaC reconstruction are required.
Security	Public, Internal, Confidential, Restricted and Regulated data	Project, perimeter, table, row and column controls are layered.
Operating model	Central platform plus federated domain product teams	Ownership and support are explicit for every product.
Delivery	Terraform and CI/CD are mandatory for platform and schema changes	Console-only configuration is noncompliant for production.

Executive Decisions

Decision	Recommended position	Executive owner
Primary analytical platform	BigQuery native tables for the majority of curated analytical data	CDO and CTO
Open lakehouse	Use Iceberg selectively for multi-engine interoperability, not as a universal default	Chief Data Architect
Ownership	Domain-owned data products on a centrally operated platform	CDO and business data owners
Regional model	Regional data planes aligned to residency, with centrally governed metadata and standards	CIO, Privacy and Security
Real-time scope	Streaming only where decision latency justifies complexity and cost	Business capability owners

Decision	Recommended position	Executive owner
Access model	Group and attribute-based access through product interfaces, not direct ad hoc table grants	Security and Governance
Cost model	Showback from day one and chargeback for production domains after stabilization	CFO and FinOps

1. Data-Platform Objectives

The platform is an enterprise product that enables business domains to create and operate trustworthy data products. It is not a collection of independent Google Cloud services and it is not a centralized team that builds every pipeline. The objectives below are measurable architectural outcomes.

ID	Objective	Architecture intent	Success measure
OBJ-01	Trusted data products	Publish certified products with named owners, contracts, lineage, quality and SLOs.	90% of priority analytical consumption uses certified products within 24 months.
OBJ-02	Elastic scale	Support petabyte storage and burst processing without fixed cluster planning for every workload.	Ingestion and analytical workloads pass 2x projected peak tests.
OBJ-03	Latency choice	Provide batch, micro-batch, CDC and streaming patterns selected by business latency.	Every feed has an approved latency class and cost justification.
OBJ-04	Security by design	Protect data according to sensitivity, residency and purpose.	100% of production assets classified; no unresolved critical access findings.
OBJ-05	Interoperability	Use open files and Iceberg where engine portability is required.	Approved multi-engine workloads avoid duplicate authoritative copies.
OBJ-06	Data quality	Detect quality issues before consumers receive incorrect data.	Tier 0 and 1 products meet defined quality thresholds and block publication on critical failure.
OBJ-07	Discoverability	Make products, definitions and owners searchable.	Priority assets are discoverable in Knowledge Catalog with complete mandatory metadata.
OBJ-08	Operational reliability	Operate pipelines and products using SLOs, runbooks and error budgets.	Tier 0/1 services publish SLOs and pass operational readiness review.
OBJ-09	Cost accountability	Allocate platform cost to domains and products.	At least 95% of data-platform spend is attributable by domain, product and environment.
OBJ-10	AI readiness	Expose governed, high-quality and semantically rich data to approved AI platforms.	AI use cases consume certified product interfaces and record lineage and purpose.

1.1 Design Outcomes

A successful implementation creates a stable set of paved roads. A source team can onboard a feed through a standard intake process; a domain team can build a product using reusable infrastructure and pipeline templates; a consumer can discover and request access without learning internal project topology; and operations can determine freshness, correctness, lineage, cost and owner from one control surface. Custom engineering remains possible, but it is an exception with explicit ownership and support consequences.

The platform must also reduce the number of authoritative copies. Raw source evidence is retained once per legal and operational requirement. Transformations create managed layers only when they add contract, quality, performance or access value. Export copies are time-bound and monitored. Multi-engine access uses BigLake and Iceberg when it avoids duplication without sacrificing governance.

1.2 Scope Boundaries

In scope	Out of scope or separate volume
Enterprise source onboarding, ingestion, storage, governance, access, sharing and operational standards	Detailed Beam, Spark, Airflow and Dataform coding standards - Volume 04
Logical and physical data layers	Enterprise semantic modeling and dashboard engineering - analytics volume
Data-product lifecycle and certification	Detailed domain decomposition and mesh operating model - Volume 05
Fine-grained data access and data encryption	Complete landing-zone and network implementation - Volume 02

In scope	Out of scope or separate volume
Platform SLOs and monitoring controls	Full SRE, incident and capacity processes - Volume 10
Sample dataset, table and access definitions	End-to-end industry implementations - Volume 12

2. Enterprise Data-Platform Capabilities

Capabilities are stable business and technology outcomes. Services are replaceable implementation choices. This distinction prevents the architecture from becoming a product inventory and provides a traceable basis for service selection.

ID	Capability	Definition	Primary implementation
CAP-01	Source onboarding	Discover owner, contract, sensitivity, latency, volume and recovery requirements before connection.	Intake workflow, source registry, approvals
CAP-02	Managed transfer	Move large files and historical data with integrity and restartability.	Storage Transfer Service, transfer appliances where justified
CAP-03	Batch ingestion	Load scheduled extracts with manifests, reconciliation and deterministic reruns.	Cloud Storage, Dataflow, BigQuery load jobs
CAP-04	Event ingestion	Decouple producers and consumers with versioned events and replay.	Pub/Sub, schemas, archive subscription
CAP-05	CDC	Capture committed database changes while preserving order and recoverability.	Datastream and merge framework
CAP-06	API integration	Ingest from rate-limited and authenticated services.	Cloud Run, Workflows, Apigee, Secret Manager
CAP-07	Open lake storage	Persist immutable and interoperable files.	Cloud Storage, Parquet, Avro, Iceberg
CAP-08	Analytical warehouse	Provide governed SQL analytics at enterprise scale.	BigQuery
CAP-09	Operational serving	Serve low-latency key-value, global relational and PostgreSQL-compatible use cases.	Bigtable, Spanner, AlloyDB
CAP-10	Metadata and discovery	Inventory assets and connect technical context to business meaning.	Knowledge Catalog
CAP-11	Lineage and impact	Trace origin, transformation, destination and consumer impact.	Knowledge Catalog lineage plus custom events
CAP-12	Data quality	Profile, validate, score, quarantine and remediate.	Knowledge Catalog automatic data quality and pipeline gates
CAP-13	Contract management	Version schema, semantics, SLO and compatibility expectations.	Contract repository, schema registry, CI validation
CAP-14	Fine-grained access	Restrict datasets, rows, columns and visible values.	IAM, authorized views, row policies, policy tags, masking
CAP-15	Data sharing	Publish governed datasets and streams across organizational boundaries.	BigQuery sharing and controlled APIs
CAP-16	Lifecycle management	Retain, archive, hold, delete and evidence disposition.	Cloud Storage retention/holds, BigQuery lifecycle, workflow
CAP-17	Observability	Measure freshness, quality, lineage completeness, pipeline health and cost.	Cloud Monitoring, Logging, product telemetry
CAP-18	FinOps	Allocate, forecast and optimize by domain and product.	Billing export, labels, reservations and unit metrics

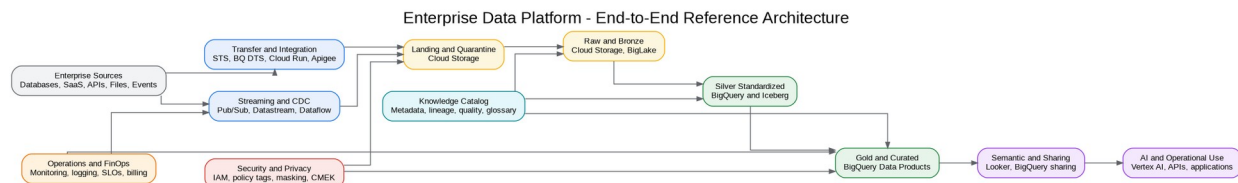
2.1 Capability Maturity Model

Level	Description	Evidence
1 - Ad hoc	Teams build independent pipelines and grant direct access.	Manual scripts, undocumented datasets and reactive support.

Level	Description	Evidence
2 - Standardized	Common templates and layer definitions exist.	Reusable modules, naming standards and basic monitoring.
3 - Governed	Ownership, contracts, catalog, access workflow and quality gates are enforced.	Certified products, automated policies and audit evidence.
4 - Productized	Domains operate data products against SLO, cost and consumer commitments.	Product scorecards, showback and versioned interfaces.
5 - Optimizing	Telemetry drives automated tuning, deprecation and policy improvement.	Predictive capacity, anomaly detection and continuous control validation.

2.2 Central Platform and Domain Responsibilities

Capability	Central platform	Domain team
Ingestion frameworks	Build and operate reusable templates, connectors, DLQ and monitoring patterns.	Configure source-specific contract and own source reconciliation.
Storage	Provide modules, lifecycle defaults and security controls.	Select layer and retention within policy; own product datasets.
Governance	Operate catalog, glossary standards, classification and certification workflow.	Supply business definitions, owner, steward and product metadata.
Quality	Provide rule framework, execution platform and scorecards.	Define thresholds, remediate failures and accept exceptions.
Access	Provide group, policy-tag, row-policy and request workflow patterns.	Approve purpose-based consumer access.
Operations	Provide telemetry, platform SLOs and escalation paths.	Own product SLO, runbook and business impact.
FinOps	Provide allocation model, budgets and recommendations.	Own product budget and act on optimization findings.



End-to-end enterprise data-platform reference architecture

Figure: End-to-end enterprise data-platform reference architecture

3. Source-System Architecture and Data-Source Classification

Source onboarding begins with architecture and risk discovery, not connectivity. The source registry records the authoritative owner, interface, legal basis, sensitivity, expected volume, change pattern, latency, retention, replay method, maintenance windows and recovery requirements. A source is not production-ready until its control and reconciliation model is approved.

3.1 Source-System Architecture

Sources are grouped by control characteristics rather than vendor alone. Systems of record require reconciliation and clear extraction boundaries. Event producers require schema ownership and replay. SaaS sources require awareness of provider limits, API changes and historical extract behavior. Mainframes require record-format and encoding expertise. Cross-cloud sources add egress, identity and residency concerns. Partner sources add contractual and nonrepudiation controls.

The platform never assumes that a successful API call or transferred file proves completeness. Each interface has a control total, sequence, watermark, source snapshot identifier or event offset that supports independent reconciliation. Where the source cannot provide such evidence, the data owner accepts a documented residual risk and defines compensating checks.

Source class	Examples	Default integration	Primary risks
Operational relational	Oracle, SQL Server, PostgreSQL, MySQL, DB2	CDC plus periodic reconciliation	Transaction order, schema changes, source load
Mainframe	DB2 z/OS, VSAM, IMS, flat files	Batch unload or approved log-based capture	EBCDIC, packed decimal, copybooks, batch windows
ERP and enterprise apps	SAP, Salesforce, Workday, ServiceNow	Managed extract, API or supported connector	Vendor limits, semantic complexity, delta behavior
Event platforms	Kafka, application events, IoT brokers	Pub/Sub or Dataflow connector	Ordering, duplicate delivery, schema compatibility
Files and objects	SFTP, NFS, object stores, managed file transfer	Manifest-based object ingestion	Partial files, duplicate delivery, malware
APIs	REST, SOAP, GraphQL	Cloud Run/Workflows with managed secrets	Rate limits, pagination, replay and provider changes
Cross-cloud	Amazon S3, Azure Data Lake, cloud warehouses	Managed transfer, direct query or replication	Egress, identity, regionality and consistency
Partners	B2B exchange, regulators, suppliers	Managed file/API, signed manifests and isolated landing	Contract, provenance, nonrepudiation and data minimization
Unstructured	Documents, images, audio, logs	Object storage and metadata extraction	Sensitive content, file types, AI processing rights

3.2 Data-Source Classification Model

Dimension	Values	Design use
Authority	System of record / derived / external / reference	Determines reconciliation and conflict resolution.
Change pattern	Append / update / delete / snapshot / event	Determines CDC, merge and history design.
Latency	Real time / near real time / hourly / daily / periodic	Selects streaming, micro-batch or batch.
Volume	Small / medium / large / extreme	Drives transfer, partition and quota design.
Sensitivity	Public / Internal / Confidential / Restricted / Regulated	Drives projects, perimeters, masking and approval.
Residency	Global / AMER / EMEA / APAC / country-restricted	Determines region, key and sharing boundaries.
Schema stability	Stable / additive / frequent breaking changes	Determines contract strictness and compatibility mode.
Replayability	Source replay / event archive / snapshot only / none	Determines recovery and evidence requirements.
Availability	Tier 0 / 1 / 2 / 3	Determines SLO and operational response.
Ownership	Named business and technical owners	Determines approvals, stewardship and support.

3.3 Source Intake Gate

Identify business owner, technical owner, steward and consumer use case.

Classify sensitivity, residency, retention, legal basis and data-subject obligations.

Record source data model, interface, supported extraction windows and provider limits.

Measure current and peak volume, row size, event rate and change rate.

Select latency class and justify why slower processing is insufficient.

Define schema contract, compatibility rules and deprecation notice.

Define source and target control totals, watermarks and reconciliation tolerance.

Define replay, backfill, deletion and historical reload behavior.

Complete threat assessment, secrets, network path and service-perimeter review.

Approve SLO, runbook, cost owner and production support model.

4. Ingestion and Data-Transfer Architecture

The ingestion plane isolates source-specific variability from domain transformations. It validates transport integrity, contract compatibility, security classification and replay evidence before data enters governed layers. Ingestion is not the place to embed complex business logic. Source-aligned normalization is allowed; cross-domain business rules belong in the standardization and product layers.

4.1 Batch-Ingestion Architecture

Use batch when the business decision can wait for a scheduled boundary and the source can provide a complete extract or stable incremental watermark. Batch is the default because it is simpler to reconcile, replay and operate than continuous processing. A scheduler starts an extract or waits for a manifest. Objects arrive in a write-only landing prefix. Validation verifies naming, checksum, encryption, source window, schema and control totals. Accepted objects are copied or promoted to immutable raw storage. Processing writes source-aligned bronze data and records reconciliation metrics before publication.

Design aspect	Enterprise standard
Primary services	Storage Transfer Service, Cloud Storage, BigQuery load jobs, Dataflow batch, Workflows or Cloud Composer, Cloud Monitoring
Idempotency and replay	Use source batch ID plus object checksum. Reprocessing a batch writes to a staging target and atomically replaces or merges the intended partition.
Failure handling	Invalid files move to quarantine with error metadata; partial batches remain unpublished; retries preserve batch ID; operational alerts include source window and failing object.
Security	Dedicated transfer identity, bucket-level uniform access, private connectivity where required, malware inspection for external files, CMEK for restricted classes.
Cost optimization	Prefer load jobs and compressed columnar formats; avoid many small files; schedule heavy transformations outside contention windows.
Common pitfalls	Using arrival time as business time; publishing before reconciliation; overwriting raw evidence; accepting silent schema drift.

Implementation sequence

Register the source and approve owner, classification, latency and recovery requirements.

Provision connectivity, service identity, secret, landing target, schema contract and monitoring through reusable infrastructure modules.

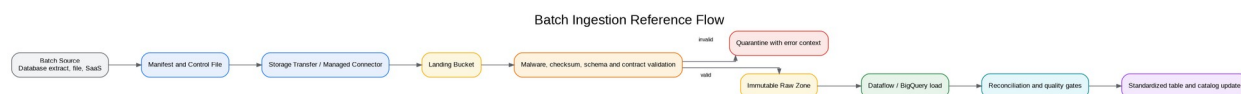
Execute a representative volume and failure test, including duplicate, late, malformed and replay scenarios.

Reconcile source and target independently and retain evidence with the ingestion ledger.

Promote the feed only after security, quality, operational and cost acceptance criteria pass.

Acceptance evidence

Evidence	Minimum acceptance
Contract	Versioned schema, semantics, owner and compatibility rule approved.
Integrity	Checksum, sequence, watermark or control total proves transfer completeness.
Recovery	Demonstrated replay or backfill without duplicate business records.
Observability	Freshness, failure, backlog and volume telemetry visible to owner.
Security	Identity, network, encryption and data-classification controls tested.
Cost	Baseline unit cost and expected monthly range recorded.



Batch-ingestion reference architecture

Figure: Batch-ingestion reference architecture

4.2 Streaming-Ingestion Architecture

Use streaming when a business action, operational alert or customer experience requires event availability in seconds or a few minutes. Streaming is not justified only because the source emits events. Producers publish versioned events to Pub/Sub or an approved connector. A Dataflow pipeline validates schema, event time and identity, deduplicates where required, enriches from bounded reference state and writes to BigQuery through the Storage Write API or to Bigtable for low-latency serving. A separate archive path preserves replay evidence.

Design aspect	Enterprise standard
Primary services	Pub/Sub, Pub/Sub schemas, Dataflow, BigQuery Storage Write API, Bigtable, Cloud Storage replay archive
Idempotency and replay	Use a stable event ID and producer sequence. Exactly-once write semantics can be used with application-created BigQuery streams and offsets; business idempotency remains necessary across retries and downstream effects.
Failure handling	Malformed events go to a dead-letter topic; backlog and watermark alerts protect freshness; poison messages have bounded retry; replay uses retained messages or archive objects.
Security	Per-producer service identities, topic-level publish permissions, schema validation, service perimeter assessment, encryption and restricted subscriber identities.
Cost optimization	Control message size and retention; use at-least-once when duplicates can be cheaply resolved; autoscale Dataflow within tested worker limits.
Common pitfalls	Assuming ordering across all events; no replay store; embedding consumer-specific logic in shared topics; unbounded state.

Implementation sequence

Register the source and approve owner, classification, latency and recovery requirements.

Provision connectivity, service identity, secret, landing target, schema contract and monitoring through reusable infrastructure modules.

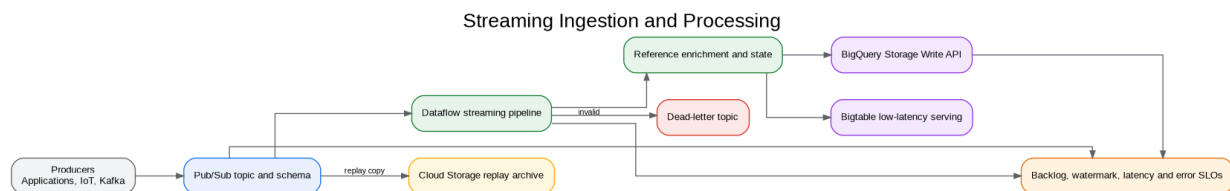
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Security	Identity, network, encryption and data-classification controls tested.
Cost	Baseline unit cost and expected monthly range recorded.



Streaming-ingestion reference architecture

Figure: Streaming-ingestion reference architecture

4.3 CDC Architecture

Use change data capture for supported operational databases when downstream consumers need committed changes more frequently than scheduled extracts and the source can support log-based capture. Datastream reads database transaction logs and writes changes to a supported destination. A landing or staging representation preserves operation type, source timestamp and sequence. A merge pipeline applies inserts, updates and deletes into bronze and silver tables while maintaining history and reconciliation.

Design aspect	Enterprise standard
Primary services	Datastream, Cloud Storage or BigQuery destination, Dataflow or BigQuery SQL merge, Secret Manager, Cloud Monitoring
Idempotency and replay	Persist source log position and primary key. Merge based on operation sequence and source commit time. Replays must not create duplicate current records or corrupt history.
Failure handling	Monitor stream state and source lag. Schema changes are evaluated against contract compatibility. Gaps trigger controlled backfill from a consistent source snapshot.
Security	Private connectivity where supported, minimal database account, secret rotation, source allowlist, per-stream service account and CMEK where required.
Cost optimization	Filter unused schemas and tables; avoid capturing columns with no consumer; tune merge cadence to balance freshness and query cost.
Common pitfalls	No initial snapshot reconciliation; missing primary keys; relying on arrival order; applying destructive schema changes automatically.

Implementation sequence

Register the source and approve owner, classification, latency and recovery requirements.

Provision connectivity, service identity, secret, landing target, schema contract and monitoring through reusable infrastructure modules.

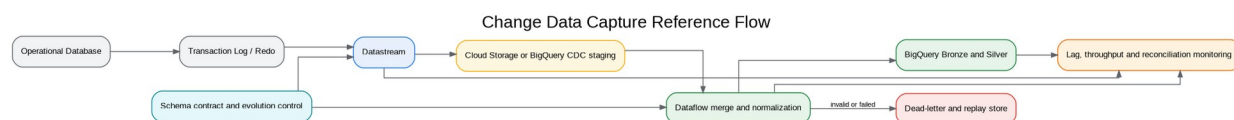
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Acceptance evidence

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Contract	Versioned schema, semantics, owner and compatibility rule approved.
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Recovery	Demonstrated replay or backfill without duplicate business records.
Observability	Freshness, failure, backlog and volume telemetry visible to owner.
Security	Identity, network, encryption and data-classification controls tested.
Cost	Baseline unit cost and expected monthly range recorded.



Change-data-capture reference architecture

Figure: Change-data-capture reference architecture

4.4 API-Ingestion Architecture

Use API ingestion where the provider exposes business objects or events only through REST, SOAP or GraphQL and no managed connector meets control requirements. A Cloud Run service or Workflows definition retrieves credentials from Secret Manager, calls the API using bounded concurrency, handles pagination and rate limits, persists raw responses and control metadata, and publishes a completion event. Parsing and business normalization occur downstream.

Design aspect	Enterprise standard
Primary services	Cloud Run, Workflows, Apigee where enterprise API policy is needed, Secret Manager, Cloud Scheduler, Pub/Sub, Cloud Storage
Idempotency and replay	Use provider object ID, update timestamp and request window. Store continuation tokens and response hashes. Reruns use the same extraction window and merge semantics.
Failure handling	Apply exponential backoff with jitter, provider-specific retryable codes, circuit breaking and a persisted checkpoint. Do not retry non-idempotent calls without a provider idempotency key.
Security	Outbound egress control, short-lived identity, managed secrets, certificate validation, payload minimization and redaction of secrets from logs.
Cost optimization	Prefer incremental endpoints, compression and bulk APIs. Bound parallelism to prevent rate-limit retries from multiplying cost.
Common pitfalls	Logging credentials or full sensitive payloads; losing pagination tokens; assuming API response order; no provider-version monitoring.

Implementation sequence

Register the source and approve owner, classification, latency and recovery requirements.

Provision connectivity, service identity, secret, landing target, schema contract and monitoring through reusable infrastructure modules.

Execute a representative volume and failure test, including duplicate, late, malformed and replay scenarios.

Reconcile source and target independently and retain evidence with the ingestion ledger.

Promote the feed only after security, quality, operational and cost acceptance criteria pass.

Acceptance evidence

Evidence	Minimum acceptance
Contract	Versioned schema, semantics, owner and compatibility rule approved.
Integrity	Checksum, sequence, watermark or control total proves transfer completeness.
Recovery	Demonstrated replay or backfill without duplicate business records.
Observability	Freshness, failure, backlog and volume telemetry visible to owner.
Security	Identity, network, encryption and data-classification controls tested.
Cost	Baseline unit cost and expected monthly range recorded.

4.5 SaaS-Ingestion Architecture

Use provider-supported transfer services or vetted connectors when they preserve source semantics, incremental extraction and operational evidence. Custom code is the fallback, not the first choice. A connector or BigQuery Data Transfer Service schedule lands provider data in a restricted dataset. A source adapter validates expected tables, extract timestamps and row counts, then copies or transforms into enterprise bronze with the source contract attached.

Design aspect	Enterprise standard
Primary services	BigQuery Data Transfer Service, supported third-party connector, Cloud Run custom adapter, Secret Manager, BigQuery
Idempotency and replay	Provider extract ID and target partition are the replay keys. Maintain an extraction ledger so late and corrected provider records can be reapplied.
Failure handling	Alert on schedule failure, row-count anomaly, provider schema change and missing objects. Retain provider extracts long enough to support correction.
Security	OAuth or service credentials in Secret Manager, least-privilege provider role, regional review and vendor risk assessment.
Cost optimization	Avoid full reloads for high-volume SaaS systems; remove unused objects; schedule outside provider peak and BigQuery contention windows.
Common pitfalls	Treating provider-managed fields as stable; unclear source ownership; no

Design aspect	Enterprise standard
	deletion handling; no contract around formula fields.

Implementation sequence

Register the source and approve owner, classification, latency and recovery requirements.

Provision connectivity, service identity, secret, landing target, schema contract and monitoring through reusable infrastructure modules.

Execute a representative volume and failure test, including duplicate, late, malformed and replay scenarios.

Reconcile source and target independently and retain evidence with the ingestion ledger.

Promote the feed only after security, quality, operational and cost acceptance criteria pass.

Acceptance evidence

Evidence	Minimum acceptance
Contract	Versioned schema, semantics, owner and compatibility rule approved.
Integrity	Checksum, sequence, watermark or control total proves transfer completeness.
Recovery	Demonstrated replay or backfill without duplicate business records.
Observability	Freshness, failure, backlog and volume telemetry visible to owner.
Security	Identity, network, encryption and data-classification controls tested.
Cost	Baseline unit cost and expected monthly range recorded.

4.6 File-Ingestion Architecture

Use file ingestion for bounded objects delivered through managed transfer, SFTP gateways, partner exchanges or application exports. The file protocol must include a completion signal. A file is written under a temporary name or prefix, followed by an atomic rename or manifest. Event notification triggers validation only after completion. The validator checks extension, MIME type, size, checksum, compression, encryption, schema and record count before moving data to raw or quarantine.

Design aspect	Enterprise standard
Primary services	Cloud Storage, Storage Transfer Service, Eventarc or Pub/Sub notification, Cloud Run validation, Dataflow or BigQuery load
Idempotency and replay	Object generation, checksum and source batch ID form the identity. Duplicate objects are recorded and ignored or versioned according to contract.
Failure handling	Never parse a partially uploaded file. Quarantine unsupported types, decompression bombs, schema mismatches and control-total failures.
Security	Uniform bucket-level access, managed upload identity, object scanning for untrusted senders, retention and signed URL controls where needed.
Cost optimization	Use Parquet or Avro for high-volume analytical files; compact small files; lifecycle landing copies promptly after acceptance.
Common pitfalls	Filename-only control; no manifest; uncontrolled ZIP archives; allowing partners to list or read other partners' objects.

Implementation sequence

Register the source and approve owner, classification, latency and recovery requirements.

Provision connectivity, service identity, secret, landing target, schema contract and monitoring through reusable infrastructure modules.

Execute a representative volume and failure test, including duplicate, late, malformed and replay scenarios.

Reconcile source and target independently and retain evidence with the ingestion ledger.

Promote the feed only after security, quality, operational and cost acceptance criteria pass.

Acceptance evidence

Evidence	Minimum acceptance
Contract	Versioned schema, semantics, owner and compatibility rule approved.
Integrity	Checksum, sequence, watermark or control total proves transfer completeness.
Recovery	Demonstrated replay or backfill without duplicate business records.
Observability	Freshness, failure, backlog and volume telemetry visible to owner.
Security	Identity, network, encryption and data-classification controls tested.
Cost	Baseline unit cost and expected monthly range recorded.

4.7 Mainframe-Ingestion Architecture

Use batch unload as the default for mainframe datasets unless a supported, economically justified log-based pattern is available. Protect source batch windows and preserve copybook semantics. Mainframe jobs create consistent extracts with copybook/version metadata and control totals. Managed file transfer moves encrypted data to a dedicated landing zone. Conversion handles EBCDIC, packed decimals, variable records and signed fields before raw preservation and bronze normalization.

Design aspect	Enterprise standard
Primary services	Enterprise scheduler, managed file transfer, Cloud Storage, Dataflow custom transforms, BigQuery, Secret Manager
Idempotency and replay	Business date, dataset name, generation number and checksum identify the extract. Replacement requires an explicit supersession record.
Failure handling	Reject copybook mismatches, truncated records and control-total differences. Maintain sample records and conversion test cases for every format version.
Security	Dedicated transfer channel, encryption, restricted projects, tightly controlled copybook repository and no sensitive record logging.
Cost optimization	Compress large fixed-width extracts; parallelize conversion by deterministic record boundaries; avoid full daily unloads when stable incremental keys exist.
Common pitfalls	Incorrect numeric conversion; character-set assumptions; losing source generation identifiers; changing copybook without versioning.

Implementation sequence

Register the source and approve owner, classification, latency and recovery requirements.

Provision connectivity, service identity, secret, landing target, schema contract and monitoring through reusable infrastructure modules.

Execute a representative volume and failure test, including duplicate, late, malformed and replay scenarios.

Reconcile source and target independently and retain evidence with the ingestion ledger.

Promote the feed only after security, quality, operational and cost acceptance criteria pass.

Acceptance evidence

Evidence	Minimum acceptance
Contract	Versioned schema, semantics, owner and compatibility rule approved.
Integrity	Checksum, sequence, watermark or control total proves transfer completeness.
Recovery	Demonstrated replay or backfill without duplicate business records.
Observability	Freshness, failure, backlog and volume telemetry visible to owner.
Security	Identity, network, encryption and data-classification controls tested.
Cost	Baseline unit cost and expected monthly range recorded.

4.8 IoT-Ingestion Architecture

Use an edge-to-cloud event pattern for telemetry where devices are intermittently connected, events can arrive late and device identity must be verified. An approved device or gateway platform authenticates devices, buffers when offline and forwards telemetry to Pub/Sub. Dataflow validates device identity, event time and units, handles late events with watermarks, and writes time-series analytical data to BigQuery and operational aggregates to Bigtable where needed.

Design aspect	Enterprise standard
Primary services	Device gateway or partner IoT platform, Pub/Sub, Dataflow, BigQuery, Bigtable, Cloud Storage
Idempotency and replay	Device ID, event timestamp and sequence number define uniqueness. Out-of-order events update windowed aggregates only within an approved lateness boundary.
Failure handling	Dead-letter invalid devices and malformed payloads; isolate noisy devices; monitor per-device and per-site gaps rather than only global throughput.
Security	Mutual authentication at the device platform, device credential rotation, topic isolation, payload minimization and geographic routing.
Cost optimization	Aggregate at the edge when lossless raw telemetry is not required; use compact schemas; set retention based on use case rather than indefinite default.
Common pitfalls	No device inventory; unbounded cardinality in metrics; accepting client clock blindly; retaining high-frequency data without business value.

Implementation sequence

Register the source and approve owner, classification, latency and recovery requirements.

Provision connectivity, service identity, secret, landing target, schema contract and monitoring through reusable infrastructure modules.

Execute a representative volume and failure test, including duplicate, late, malformed and replay scenarios.

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Observability	Freshness, failure, backlog and volume telemetry visible to owner.
Security	Identity, network, encryption and data-classification controls tested.
Cost	Baseline unit cost and expected monthly range recorded.

4.9 Cross-Cloud Ingestion Architecture

Use managed transfer or governed federation based on query frequency, data gravity, sovereignty, egress and recovery. A cross-cloud query is not automatically cheaper or simpler than replication. For repeat analytics, managed transfer copies versioned objects or extracts into the regional data plane. For limited discovery, approved federation may query external storage. Cross-cloud credentials use workload identity federation where supported. Control totals and object inventories prove completeness.

Design aspect	Enterprise standard
Primary services	Storage Transfer Service, BigLake external tables, BigQuery Omni where appropriate and validated, Workload Identity Federation, Cloud Storage
Idempotency and replay	Object version, ETag or provider generation and transfer job ID identify copies. Deletions and replacements follow an explicit mirror policy.
Failure handling	Monitor provider throttling, egress errors and partial inventories. Recovery

Design aspect	Enterprise standard
	can restart by object generation rather than full recopy.
Security	No long-lived cloud keys, explicit egress allowlists, residency review, encryption, isolated projects and contractually approved external access.
Cost optimization	Model source egress, inter-region transfer, repeated query scans and duplicate retention. Move stable high-use data close to consumers.
Common pitfalls	Hidden egress cost; unsupported fine-grained controls; ambiguous authoritative copy; direct analyst credentials to external cloud.

Implementation sequence

Register the source and approve owner, classification, latency and recovery requirements.

Provision connectivity, service identity, secret, landing target, schema contract and monitoring through reusable infrastructure modules.

Execute a representative volume and failure test, including duplicate, late, malformed and replay scenarios.

Reconcile source and target independently and retain evidence with the ingestion ledger.

Promote the feed only after security, quality, operational and cost acceptance criteria pass.

Acceptance evidence

Evidence	Minimum acceptance
Contract	Versioned schema, semantics, owner and compatibility rule approved.
Integrity	Checksum, sequence, watermark or control total proves transfer completeness.
Recovery	Demonstrated replay or backfill without duplicate business records.
Observability	Freshness, failure, backlog and volume telemetry visible to owner.
Security	Identity, network, encryption and data-classification controls tested.
Cost	Baseline unit cost and expected monthly range recorded.

4.10 Partner-Data Ingestion Architecture

Use isolated landing and explicit contractual controls for data supplied by customers, suppliers, regulators and data vendors. Each partner receives a dedicated identity and prefix, endpoint or API product. Signed manifests, checksums and delivery windows are verified. The platform applies malware, schema, license and quality checks before data can enter raw storage.

Design aspect	Enterprise standard
Primary services	Cloud Storage, Apigee or Cloud Run, Private Service Connect where applicable, Cloud KMS, Pub/Sub, validation service
Idempotency and replay	Partner delivery ID and content hash identify the submission. Corrections are new versions that reference the superseded delivery.
Failure handling	Quarantine invalid or late deliveries; notify the partner through the agreed channel; retain nonrepudiation evidence and processing outcome.
Security	Partner isolation, least privilege, key rotation, allowlists or private connectivity, payload classification and contractual use restrictions.
Cost optimization	Use shared validation services but separate storage and billing attribution. Define who pays for reprocessing caused by partner defects.
Common pitfalls	Shared partner folder; no contract for corrections; using data outside licensed purpose; no offboarding and credential revocation.

Implementation sequence

Register the source and approve owner, classification, latency and recovery requirements.

Provision connectivity, service identity, secret, landing target, schema contract and monitoring through reusable infrastructure modules.

Execute a representative volume and failure test, including duplicate, late, malformed and replay scenarios.

Reconcile source and target independently and retain evidence with the ingestion ledger.

Promote the feed only after security, quality, operational and cost acceptance criteria pass.

Acceptance evidence

Evidence	Minimum acceptance
Contract	Versioned schema, semantics, owner and compatibility rule approved.
Integrity	Checksum, sequence, watermark or control total proves transfer completeness.
Recovery	Demonstrated replay or backfill without duplicate business records.
Observability	Freshness, failure, backlog and volume telemetry visible to owner.
Security	Identity, network, encryption and data-classification controls tested.
Cost	Baseline unit cost and expected monthly range recorded.

4.11 Historical Data Transfer Architecture

Use a staged migration pattern for terabyte- and petabyte-scale historical data. The objective is verifiable transfer with minimal source disruption, not the fastest possible copy at any cost. Inventory objects and tables, classify data, select online transfer or appliance, seed bulk history, run incremental deltas, reconcile independently, freeze the source window, apply final delta and certify the target.

Design aspect	Enterprise standard
Primary services	Storage Transfer Service, Transfer Appliance when justified, Database Migration Service for appropriate databases, Cloud Storage, BigQuery load
Idempotency and replay	Inventory manifests, source checksums, object generations and target reconciliation records are retained for every wave.
Failure handling	Retry at object or partition granularity; do not restart entire waves; maintain rollback until consumers accept the target.
Security	Chain of custody, encryption, appliance handling procedures, transfer identities, data residency and temporary staging deletion.
Cost optimization	Compare network, appliance, egress, source load, compression and parallelism. Archive data may move directly to lower-cost storage classes.
Common pitfalls	No inventory baseline; combining transformation with initial transfer; deleting source before consumer acceptance; underestimating small-file overhead.

Implementation sequence

Register the source and approve owner, classification, latency and recovery requirements.

Provision connectivity, service identity, secret, landing target, schema contract and monitoring through reusable infrastructure modules.

Execute a representative volume and failure test, including duplicate, late, malformed and replay scenarios.

Reconcile source and target independently and retain evidence with the ingestion ledger.

Promote the feed only after security, quality, operational and cost acceptance criteria pass.

Acceptance evidence

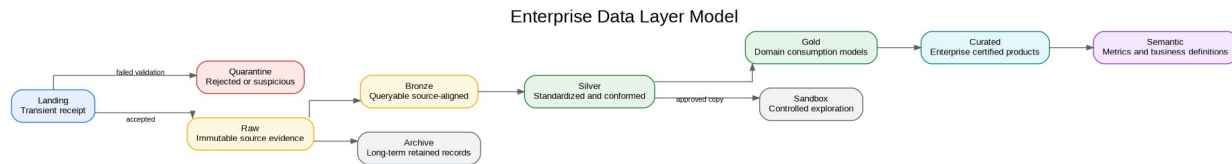
Evidence	Minimum acceptance
Contract	Versioned schema, semantics, owner and compatibility rule approved.
Integrity	Checksum, sequence, watermark or control total proves transfer completeness.
Recovery	Demonstrated replay or backfill without duplicate business records.
Observability	Freshness, failure, backlog and volume telemetry visible to owner.
Security	Identity, network, encryption and data-classification controls tested.
Cost	Baseline unit cost and expected monthly range recorded.

4.12 Ingestion Pattern Selection Matrix

Requirement	Batch	CDC	Streaming events	API/SaaS	Managed transfer
Latency	Minutes to days	Seconds to minutes	Sub-second to minutes	Minutes to hours	Hours to days
Source impact	Scheduled extract load	Transaction-log load	Producer integration	Provider rate limits	Object inventory and copy
Replay	Batch ID and raw files	Log position plus snapshots	Message retention/archive	Window and checkpoint	Object generation
Schema change	At batch boundary	Continuous and potentially disruptive	Contract compatibility	Provider controlled	File contract
Best fit	Reporting and periodic products	Operational change propagation	Time-sensitive event processing	External applications	Historical or cross-cloud movement
Avoid when	Immediate action is required	No stable key/log support	Latency has no business value	Bulk export exists	Data must remain remote and low-use

5. Data-Storage and Layer Architecture

The storage model separates evidence, engineering state and consumer interfaces. Layer names are defined by contract and control, not by color alone. A dataset does not become silver or gold because of its location; it earns the designation through transformation, quality, ownership and access criteria.



Enterprise data-layer model

Figure: Enterprise data-layer model

5.1 Landing Zone

Purpose. Transient receipt area for incomplete or not-yet-validated deliveries. The reference technology is **Cloud Storage**. Typical retention is **Hours to 7 days**. Access is limited to Ingestion identity only; no consumer access.

Control area	Standard
Entry criteria	Object completion, checksum, malware and manifest checks.
Exit criteria	A downstream layer accepts the data only after contract, quality and reconciliation checks appropriate to that layer.
Metadata	Owner, source, classification, retention, schema version and lineage are mandatory.
Security	Access follows least privilege and the layer's sensitivity; raw and quarantine are more restricted than curated interfaces.
Operations	Freshness, volume, failure and storage growth are monitored.
Anti-pattern	Do not use landing as a permanent raw lake.

5.2 Quarantine Zone

Purpose. Isolation for malformed, suspicious, contract-breaking or quality-failing records. The reference technology is **Cloud Storage and restricted BigQuery error tables**. Typical retention is **30-90 days or case-specific**. Access is limited to Operations, source owner and steward.

Control area	Standard
Entry criteria	Failure reason, original identity, correlation ID and remediation status.
Exit criteria	A downstream layer accepts the data only after contract, quality and reconciliation checks appropriate to that layer.
Metadata	Owner, source, classification, retention, schema version and lineage are mandatory.
Security	Access follows least privilege and the layer's sensitivity; raw and quarantine are more restricted than curated interfaces.
Operations	Freshness, volume, failure and storage growth are monitored.
Anti-pattern	Never silently drop invalid data.

5.3 Raw Zone

Purpose. Immutable source evidence in original or lossless canonical form. The reference technology is **Cloud Storage; BigQuery only where source is inherently tabular and immutable**. Typical retention is **Legal and replay requirement**. Access is limited to Restricted engineering and audit access.

Control area	Standard
Entry criteria	Source metadata, checksum, encryption and retention.

Control area	Standard
Exit criteria	A downstream layer accepts the data only after contract, quality and reconciliation checks appropriate to that layer.
Metadata	Owner, source, classification, retention, schema version and lineage are mandatory.
Security	Access follows least privilege and the layer's sensitivity; raw and quarantine are more restricted than curated interfaces.
Operations	Freshness, volume, failure and storage growth are monitored.
Anti-pattern	No destructive updates; corrections create new versions.

5.4 Bronze Layer

Purpose. Queryable source-aligned data with technical normalization and ingestion metadata. The reference technology is **BigQuery native or Iceberg table**. Typical retention is **Operational history requirement**. Access is limited to Domain engineers and controlled troubleshooting.

Control area	Standard
Entry criteria	Type conversion, deduplication, operation metadata and basic validity.
Exit criteria	A downstream layer accepts the data only after contract, quality and reconciliation checks appropriate to that layer.
Metadata	Owner, source, classification, retention, schema version and lineage are mandatory.
Security	Access follows least privilege and the layer's sensitivity; raw and quarantine are more restricted than curated interfaces.
Operations	Freshness, volume, failure and storage growth are monitored.
Anti-pattern	Do not embed cross-domain business definitions.

5.5 Silver Layer

Purpose. Standardized, conformed and quality-controlled domain entities. The reference technology is **BigQuery; Iceberg when multi-engine writes are required**. Typical retention is **Domain policy**. Access is limited to Domain producers and approved consumers.

Control area	Standard
Entry criteria	Canonical types, reference alignment, identity resolution and SCD policy.
Exit criteria	A downstream layer accepts the data only after contract, quality and reconciliation checks appropriate to that layer.
Metadata	Owner, source, classification, retention, schema version and lineage are mandatory.
Security	Access follows least privilege and the layer's sensitivity; raw and quarantine are more restricted than curated interfaces.
Operations	Freshness, volume, failure and storage growth are monitored.
Anti-pattern	Breaking schema changes require product versioning.

5.6 Gold Layer

Purpose. Domain-oriented aggregates and consumption models optimized for known use cases. The reference technology is **BigQuery**. Typical retention is **Business retention**. Access is limited to Named product consumers.

Control area	Standard
Entry criteria	Business calculations, performance design and product quality gates.
Exit criteria	A downstream layer accepts the data only after contract, quality and reconciliation checks appropriate to that layer.
Metadata	Owner, source, classification, retention, schema version and lineage are mandatory.

Control area	Standard
Security	Access follows least privilege and the layer's sensitivity; raw and quarantine are more restricted than curated interfaces.
Operations	Freshness, volume, failure and storage growth are monitored.
Anti-pattern	Avoid duplicating enterprise definitions across domains.

5.7 Curated Layer

Purpose. Enterprise-certified, cross-domain products and regulatory datasets. The reference technology is **BigQuery and governed views**. Typical retention is **Enterprise or regulatory policy**. Access is limited to Broad approved enterprise consumers.

Control area	Standard
Entry criteria	Governance certification, semantic review, lineage and stricter SLOs.
Exit criteria	A downstream layer accepts the data only after contract, quality and reconciliation checks appropriate to that layer.
Metadata	Owner, source, classification, retention, schema version and lineage are mandatory.
Security	Access follows least privilege and the layer's sensitivity; raw and quarantine are more restricted than curated interfaces.
Operations	Freshness, volume, failure and storage growth are monitored.
Anti-pattern	Certification is an operating commitment, not a label.

5.8 Semantic Layer

Purpose. Reusable measures, dimensions, KPI definitions and access-safe analytical interfaces. The reference technology is **Looker semantic model, BigQuery views and metrics artifacts**. Typical retention is **Version-controlled**. Access is limited to BI tools, analysts and approved applications.

Control area	Standard
Entry criteria	Metric ownership, test cases and backward compatibility.
Exit criteria	A downstream layer accepts the data only after contract, quality and reconciliation checks appropriate to that layer.
Metadata	Owner, source, classification, retention, schema version and lineage are mandatory.
Security	Access follows least privilege and the layer's sensitivity; raw and quarantine are more restricted than curated interfaces.
Operations	Freshness, volume, failure and storage growth are monitored.
Anti-pattern	Do not allow dashboards to redefine certified KPIs independently.

5.9 Sandbox Layer

Purpose. Time-bound exploration using approved copies or masked data. The reference technology is **BigQuery sandbox datasets and controlled Cloud Storage**. Typical retention is **7-90 days**. Access is limited to Named project groups; no production dependencies.

Control area	Standard
Entry criteria	Budget, expiry, restricted data rules and no inbound production contract.
Exit criteria	A downstream layer accepts the data only after contract, quality and reconciliation checks appropriate to that layer.
Metadata	Owner, source, classification, retention, schema version and lineage are mandatory.
Security	Access follows least privilege and the layer's sensitivity; raw and quarantine are more restricted than curated interfaces.
Operations	Freshness, volume, failure and storage growth are monitored.

Control area	Standard
Anti-pattern	Sandbox outputs must be productized before operational use.

5.10 Archive Layer

Purpose. Long-term, low-access records retained for compliance, recovery or historical analysis. The reference technology is **Cloud Storage Nearline/Coldline/Archive based on access pattern**. Typical retention is **Record schedule**. Access is limited to Privileged restore process.

Control area	Standard
Entry criteria	Retention lock where approved, inventory and restore tests.
Exit criteria	A downstream layer accepts the data only after contract, quality and reconciliation checks appropriate to that layer.
Metadata	Owner, source, classification, retention, schema version and lineage are mandatory.
Security	Access follows least privilege and the layer's sensitivity; raw and quarantine are more restricted than curated interfaces.
Operations	Freshness, volume, failure and storage growth are monitored.
Anti-pattern	Archive is not a substitute for a documented restore process.

5.11 Data-Layer Standards

Layer	Authoritative purpose	Mutation policy	Consumer contract	Typical optimization
Landing	Transfer handoff	Mutable only during upload; short-lived	None	Object size and transfer throughput
Quarantine	Failure evidence	Append status and remediation metadata	Operational only	Search by batch, error and owner
Raw	Source evidence	Append/version only	Source contract	Compression and lifecycle
Bronze	Queryable source alignment	Controlled merge for CDC	Technical schema	Partition by source event or ingestion date
Silver	Domain standard	Versioned merge and history	Domain entity contract	Partition and cluster for domain access
Gold	Use-case optimized	Managed rebuild or incremental update	Product interface	Materialization and clustering
Curated	Enterprise certified	Strict change governance	Enterprise product contract	Workload isolation and stable views
Semantic	Metric and meaning	Git-versioned model	KPI and semantic contract	Caching and aggregate awareness
Sandbox	Exploration	User-managed within expiry	No production contract	Budgets and expiry
Archive	Retention and recovery	Immutable under policy	Restore contract	Storage class and inventory

6. Core Data Services and Service-Selection Guidance

The platform uses a small number of strategic services, each with a defined role. The selection model prevents BigQuery from becoming an operational database, prevents Bigtable from being used for relational reporting and prevents Iceberg from being introduced without a real multi-engine requirement.

6.1 Cloud Storage Architecture

Cloud Storage is the durable object foundation for landing, quarantine, raw, replay and archive data. Buckets are regional or dual-region according to residency and recovery, use uniform bucket-level access, public access prevention and lifecycle rules. Restricted buckets use CMEK when required. Object names include domain, source, contract version, business date and batch or event identity.

Area	Guidance
Benefits and limitations	Benefits include massive scale, open-format support and independent lifecycle. Limitations include object semantics rather than database transactions, small-object overhead and the need for external metadata.
Security	Use dedicated service identities, least privilege, regional data placement, audit logging and CMEK/service perimeters where required.
Monitoring and cost	Monitor object count, bytes, request errors, lifecycle effectiveness, unexpected egress and retention-policy changes. Optimize with compression, file compaction, suitable storage classes and regional locality.
Alternatives	Select an alternative only when the workload requires different consistency, latency, openness, compatibility or operational characteristics.
Common pitfalls	Do not use bucket ACLs, mutable raw objects, globally shared catch-all buckets or irreversible retention locks without legal approval.
Production gate	Validate regional support, quotas, encryption, service-perimeter compatibility, failure behavior and recovery before approval.

6.2 BigLake Architecture

BigLake provides governed access to data in Cloud Storage and across supported external stores. Use BigLake external tables for controlled query access to open files and Apache Iceberg managed or external tables when more than one engine must operate on the same table.

Area	Guidance
Benefits and limitations	Benefits include lake and warehouse convergence, customer-owned storage and open-engine interoperability. Limitations and feature compatibility vary by table type, engine and region; teams must validate DML, governance, performance and preview status.
Security	Use dedicated service identities, least privilege, regional data placement, audit logging and CMEK/service perimeters where required.
Monitoring and cost	Monitor query scan, metadata refresh, file layout, authorization failures and cross-region access. Optimize Parquet size, partition layout and metadata maintenance.
Alternatives	Select an alternative only when the workload requires different consistency, latency, openness, compatibility or operational characteristics.
Common pitfalls	Do not introduce Iceberg only for fashion, mix unmanaged writers without concurrency testing or assume every BigQuery feature behaves identically across all external table types.
Production gate	Validate regional support, quotas, encryption, service-perimeter compatibility, failure behavior and recovery before approval.

6.3 BigQuery Architecture

BigQuery is the primary analytical warehouse and data-product serving layer. Domain projects own datasets; environment and region are explicit; reservations isolate production BI, engineering and data science where workload predictability justifies capacity.

Area	Guidance
Benefits and limitations	Benefits include serverless scale, GoogleSQL, fine-grained security, ML and sharing integration. Limitations include cost sensitivity to scan patterns, quotas, regional boundaries and differences from operational databases.
Security	Use dedicated service identities, least privilege, regional data placement, audit logging and CMEK/service perimeters where required.
Monitoring and cost	Monitor slot utilization, queued time, bytes processed, table growth, failed jobs, long-running queries, storage age and per-product cost. Optimize partition pruning, clustering, materialized results, workload assignment and query review.
Alternatives	Select an alternative only when the workload requires different consistency, latency, openness, compatibility or operational characteristics.
Common pitfalls	Do not use SELECT *, ingestion-time partitions without need, unbounded cross joins, direct production owner grants or datasets that mix unrelated security classes.
Production gate	Validate regional support, quotas, encryption, service-perimeter compatibility, failure behavior and recovery before approval.

6.4 Bigtable Use Cases

Use Bigtable for very high-throughput, low-latency key-value and wide-column workloads such as device state, time-series serving, personalization features and operational aggregates. Schema design begins with access patterns and row-key distribution.

Area	Guidance
Benefits and limitations	Benefits include horizontal scale and predictable low-latency access. It is not a general relational database; joins and arbitrary relational queries are not the design center.
Security	Use dedicated service identities, least privilege, regional data placement, audit logging and CMEK/service perimeters where required.
Monitoring and cost	Monitor per-app-profile latency, CPU, storage, hotspot indicators and replication. Optimize row keys, garbage collection, app profiles and cluster sizing.
Alternatives	Select an alternative only when the workload requires different consistency, latency, openness, compatibility or operational characteristics.
Common pitfalls	Do not use sequential hot row keys, large single rows, relational normalization or analytical full scans on serving clusters without workload isolation.
Production gate	Validate regional support, quotas, encryption, service-perimeter compatibility, failure behavior and recovery before approval.

6.5 Spanner Use Cases

Use Spanner for mission-critical relational applications that require transactional consistency with regional or global scale, strong availability and horizontal growth. It may also support graph, search or vector-enabled operational patterns where validated.

Area	Guidance
Benefits and limitations	Benefits include relational semantics, synchronous replication and scale. Limitations include a different cost and schema design model from conventional single-instance databases and the need to choose instance configuration carefully.

Area	Guidance
Security	Use dedicated service identities, least privilege, regional data placement, audit logging and CMEK/service perimeters where required.
Monitoring and cost	Monitor CPU, storage, transaction latency, lock contention, query plans and replication health. Optimize keys, interleaving where appropriate, indexes and transaction scope.
Alternatives	Select an alternative only when the workload requires different consistency, latency, openness, compatibility or operational characteristics.
Common pitfalls	Do not select Spanner only for large storage, use monotonically increasing hot keys or run unbounded analytical scans against critical transactional capacity.
Production gate	Validate regional support, quotas, encryption, service-perimeter compatibility, failure behavior and recovery before approval.

6.6 AlloyDB Use Cases

Use AlloyDB for demanding PostgreSQL-compatible transactional applications, operational marts and modernization scenarios where application compatibility and regional high availability matter more than global horizontal write scale.

Area	Guidance
Benefits and limitations	Benefits include PostgreSQL compatibility and managed enterprise operation. Limitations include regional placement, connection architecture and the responsibility to design backups, read pools and failover.
Security	Use dedicated service identities, least privilege, regional data placement, audit logging and CMEK/service perimeters where required.
Monitoring and cost	Monitor database, instance and read-pool metrics, connections, query latency, storage and backup status. Optimize indexes, connection pooling, read pools and query plans.
Alternatives	Select an alternative only when the workload requires different consistency, latency, openness, compatibility or operational characteristics.
Common pitfalls	Do not use AlloyDB as the enterprise warehouse, expose it directly to uncontrolled analyst queries or assume every PostgreSQL extension is supported without validation.
Production gate	Validate regional support, quotas, encryption, service-perimeter compatibility, failure behavior and recovery before approval.

6.7 Pub/Sub

Pub/Sub is the cloud-native event backbone for decoupled producers and consumers. Topics represent stable business event contracts; subscriptions represent consumer processing needs.

Area	Guidance
Benefits and limitations	Benefits include managed scale, independent subscriptions and replay through retention or snapshots. Delivery is at least once by default, so consumers must handle duplicates.
Security	Use dedicated service identities, least privilege, regional data placement, audit logging and CMEK/service perimeters where required.
Monitoring and cost	Monitor oldest unacked message, backlog bytes, publish errors, subscriber throughput and dead-letter volume. Optimize retention, message size and subscriber concurrency.
Alternatives	Select an alternative only when the workload requires different consistency, latency, openness, compatibility or operational characteristics.
Common pitfalls	Do not share one topic for unrelated event types, embed sensitive secrets, assume global ordering or allow unmanaged subscriber proliferation.
Production gate	Validate regional support, quotas, encryption, service-perimeter compatibility, failure behavior and recovery before approval.

6.8 Dataflow

Dataflow executes Apache Beam pipelines for batch and streaming transformations that need parallel processing, state, windows, event time or reusable connectors.

Area	Guidance
Benefits and limitations	Benefits include managed autoscaling and a common programming model. Limitations include pipeline engineering complexity, worker networking and cost if jobs are poorly bounded.
Security	Use dedicated service identities, least privilege, regional data placement, audit logging and CMEK/service perimeters where required.
Monitoring and cost	Monitor system lag, watermark, throughput, worker CPU/memory, failed bundles and autoscaling. Optimize fusion, batching, state size, worker type and Streaming Engine where appropriate.
Alternatives	Select an alternative only when the workload requires different consistency, latency, openness, compatibility or operational characteristics.
Common pitfalls	Do not place simple SQL transformations in Beam, use unbounded side inputs or omit load tests with realistic skew.
Production gate	Validate regional support, quotas, encryption, service-perimeter compatibility, failure behavior and recovery before approval.

6.9 Datastream

Datastream provides serverless CDC and replication for supported databases and destinations. It captures source changes but does not eliminate the need for target merge, history, schema and reconciliation design.

Area	Guidance
Benefits and limitations	Benefits include reduced connector operations and low-latency capture. Limitations depend on supported sources, objects, connectivity, schema behavior and regions.
Security	Use dedicated service identities, least privilege, regional data placement, audit logging and CMEK/service perimeters where required.
Monitoring and cost	Monitor stream status, source lag, errors, throughput and rejected events. Optimize included objects and merge cadence.
Alternatives	Select an alternative only when the workload requires different consistency, latency, openness, compatibility or operational characteristics.
Common pitfalls	Do not capture the entire database by default, ignore source log retention or treat CDC order as business history without keys and sequence.
Production gate	Validate regional support, quotas, encryption, service-perimeter compatibility, failure behavior and recovery before approval.

6.10 Storage Transfer Service

Storage Transfer Service manages object movement from supported sources to Cloud Storage and between buckets. It is the default for recurring or large-scale object transfer where restartability and inventory matter.

Area	Guidance
Benefits and limitations	Benefits include managed scheduling and object-level retry. Limitations include source-specific features, network and egress dependencies.
Security	Use dedicated service identities, least privilege, regional data placement, audit logging and CMEK/service perimeters where required.
Monitoring and cost	Monitor job status, bytes, object counts, failures and skipped items. Optimize parallelism, change detection and source layout.
Alternatives	Select an alternative only when the workload requires different consistency, latency, openness, compatibility or operational characteristics.
Common pitfalls	Do not use custom copy scripts for petabyte migrations without a

Area	Guidance
Production gate	compelling capability gap.
	Validate regional support, quotas, encryption, service-perimeter compatibility, failure behavior and recovery before approval.

6.11 Knowledge Catalog and Dataplex Capabilities

Knowledge Catalog is the enterprise metadata, discovery and governance plane. It inventories assets, connects technical and business context, supports glossary, lineage, profiling and quality, and provides context for data and AI. Dataplex APIs and features continue to underpin several governance operations.

Area	Guidance
Benefits and limitations	Benefits include native integration and centralized discoverability. Limitations include service-specific lineage coverage, metadata quality dependence and product evolution that must be monitored.
Security	Use dedicated service identities, least privilege, regional data placement, audit logging and CMEK/service perimeters where required.
Monitoring and cost	Monitor catalog coverage, missing mandatory metadata, scan failures, lineage gaps, quality rule outcomes and stale ownership.
Alternatives	Select an alternative only when the workload requires different consistency, latency, openness, compatibility or operational characteristics.
Common pitfalls	Do not treat automatic discovery as complete governance or allow unowned assets to become certified.
Production gate	Validate regional support, quotas, encryption, service-perimeter compatibility, failure behavior and recovery before approval.

6.12 BigQuery Sharing

BigQuery sharing, formerly Analytics Hub, publishes governed datasets and streams through exchanges and listings. Subscribers receive linked datasets rather than uncontrolled exports.

Area	Guidance
Benefits and limitations	Benefits include controlled sharing across project and organizational boundaries. Limitations include location, feature and subscriber-governance considerations.
Security	Use dedicated service identities, least privilege, regional data placement, audit logging and CMEK/service perimeters where required.
Monitoring and cost	Monitor listings, subscribers, access changes, query usage and data-product SLOs. Optimize by sharing views or products rather than broad raw datasets.
Alternatives	Select an alternative only when the workload requires different consistency, latency, openness, compatibility or operational characteristics.
Common pitfalls	Do not publish unclassified tables, grant direct owner roles to subscribers or treat a listing as a substitute for a contract.
Production gate	Validate regional support, quotas, encryption, service-perimeter compatibility, failure behavior and recovery before approval.

6.13 Technology Selection Matrix

Need	Preferred service	Conditional alternative	Decision test
Enterprise analytical SQL	BigQuery	Iceberg plus Spark for multi-engine transformations	Is governed SQL analytics the primary workload?
Immutable/open object data	Cloud Storage	None	Does the data require object retention or multi-engine files?
Open table interoperability	Apache Iceberg managed/external	BigQuery native	Must multiple engines read/write

Need	Preferred service	Conditional alternative	Decision test
High-rate event transport	tables	Existing Kafka with controlled bridge	one authoritative table?
	Pub/Sub		Are cloud-native producers/consumers and independent subscriptions required?
Stateful stream processing	Dataflow	Spark Structured Streaming	Are event-time, windows, state and autoscaling needed?
Database CDC	Datastream	Approved third-party CDC	Is the source and required behavior supported?
Low-latency wide-column serving	Bigtable	Spanner or AlloyDB	Is access key-based with very high throughput?
Global transactional relational	Spanner	Regional AlloyDB	Is strong transactional consistency across scale/regions required?
PostgreSQL-compatible transaction	AlloyDB	Cloud SQL or Spanner PostgreSQL dialect	Is PostgreSQL compatibility the main constraint?
Metadata and governance	Knowledge Catalog	Enterprise third-party catalog integration	Can native coverage satisfy governance and integration needs?
Controlled data publication	BigQuery sharing	API or governed export	Can consumers use linked datasets or stream listings?

7. Open Formats, Schema Evolution and Physical Design

Physical design is a product responsibility supported by platform standards. File and table choices are based on access path, update pattern, engine interoperability, schema evolution, retention and unit cost. No single format is optimal for every layer.

7.1 Open-Data Formats

Format	Preferred use	Strengths	Constraints
Parquet	Large analytical files and Iceberg data files	Columnar compression, predicate pushdown and broad engine support	Not a transaction format by itself; requires compaction and metadata discipline.
Avro	Event payloads, row-oriented interchange and evolving schemas	Schema embedding, compact binary and strong streaming ecosystem	Less efficient for wide analytical scans than columnar formats.
JSON	Low-volume APIs, diagnostics and human-readable exchange	Flexible and ubiquitous	Verbose, weak types and expensive at scale.
CSV	Legacy exchange only	Simple compatibility	No native types, escaping ambiguity and poor compression.
Apache Iceberg	Open table format for multi-engine lakehouse	Snapshots, schema evolution, partition evolution and transactional metadata	Operational complexity and feature compatibility must be tested.
BigQuery native	Primary curated analytical storage	Managed performance, security and integrated governance	Not an open file format; external engines use supported interfaces rather than direct files.

7.2 Apache Iceberg Considerations

Select Iceberg when the enterprise needs a common authoritative table across BigQuery and one or more open engines, when customer-owned object storage is a requirement, or when an existing Iceberg estate is being modernized. Prefer managed capabilities where they meet engine and governance needs. Define one catalog authority, supported writer engines, commit concurrency, compaction, snapshot expiration and orphan-file cleanup. Test access control and row/column policy behavior for every engine rather than assuming BigQuery-native semantics apply universally.

Do not create a parallel Iceberg copy of every BigQuery table. The additional metadata, file maintenance and engine compatibility surface must create measurable value. Curated BI products with BigQuery-only consumers generally remain native BigQuery tables. Raw and bronze open-format data can use Parquet without becoming Iceberg tables until transactional table semantics are required.

7.3 Parquet Standards

Standard	Guidance
File size	Target approximately 256 MB to 1 GB for large analytical datasets; tune to engine and workload.
Compression	Use a broadly supported codec such as Snappy or Zstandard after validation.
Schema	Use stable field names, logical types and explicit nullability.
Partitioning	Use business predicates with bounded cardinality; avoid folder-per-unique-value.
Statistics	Ensure writers produce usable column statistics and row-group metadata.
Compaction	Compact small files through managed jobs with idempotent output.
Security	Do not embed secrets or uncontrolled sensitive metadata in paths.

7.4 Avro Standards

Standard	Guidance
Schema identity	Register schema ID and compatibility mode with every event or batch contract.

Standard	Guidance
Compatibility	Backward compatible by default for shared event streams; breaking changes require a new major version/topic.
Defaults	Provide safe defaults for new optional fields.
Logical types	Use date, timestamp and decimal logical types consistently.
Union use	Avoid complex unions that create inconsistent mappings in downstream engines.
Sensitive fields	Classify in the contract and apply producer-side minimization.

7.5 Schema Evolution

Schema evolution is a governed product change. Additive nullable fields are generally backward compatible. Renames are implemented as add-populate-deprecate rather than destructive rename. Type widening requires consumer testing. Type narrowing, semantic reinterpretation and key changes are breaking. The contract repository records compatibility mode, effective date, producer notice period and migration path. CI validates source schemas, table definitions, policy tags and dependent views before promotion.

Change	Default classification	Required action
Add nullable field	Backward compatible	Update contract and tests; no major version.
Add required field without default	Breaking	New major version or staged default population.
Rename field	Breaking	Introduce new field, dual-publish, migrate and deprecate old field.
Widen numeric type	Conditionally compatible	Test consumers and update contract.
Narrow or reinterpret type	Breaking	New version and migration.
Change key or grain	Breaking semantic change	New product version and reconciliation.
Remove field	Breaking	Deprecation notice, usage evidence and major version.

7.6 BigQuery Partitioning Standards

Partition large fact and event tables by the column that most reliably bounds consumer queries, normally business event date. Require partition filters on high-cost production tables where appropriate. Use ingestion-time partitioning only when source event time is unavailable or untrustworthy. Integer-range partitioning is reserved for stable numeric ranges. Partition counts and modification patterns must remain within service limits, which are validated during design.

Decision	Standard
Partition column	Top-level DATE, TIMESTAMP or DATETIME aligned to dominant filter.
Granularity	Daily by default; hourly only for very high volume and operational need; monthly for low-volume long history.
Filter requirement	Enabled for expensive production tables and enforced in product query examples.
Late data	Write to the correct business partition; monitor late-arrival rate.
Expiration	Set only when approved by retention schedule and downstream recovery needs.
Backfill	Use partition-scoped replacement or MERGE with audit evidence.

7.7 BigQuery Clustering Standards

Cluster tables when queries repeatedly filter, aggregate or join on selective columns and partitioning alone does not sufficiently prune data. Place the most frequently filtered high-value column first. Use up to four clustering columns only when each has demonstrated benefit. Review recommendations and query history quarterly; clustering is not a one-time design decision.

Workload	Example clustering order
Customer events	customer_id, event_type, region_code

Workload	Example clustering order
Orders	customer_id, order_status, sales_channel
Finance ledger	legal_entity_id, account_id, currency_code
Telemetry	device_id, site_id, event_type
Access audit	principal, resource_type, action

8. Data Lifecycle, Retention, Archival, Legal Hold and Privacy Deletion

Lifecycle controls implement records policy, privacy obligations, recovery and cost management. Retention is derived from business purpose and legal schedule, not from storage affordability. A record may have multiple obligations; the most restrictive valid control applies until legal, regulatory and business owners approve disposition.



Data lifecycle, retention and privacy controls

Figure: Data lifecycle, retention and privacy controls

8.1 Lifecycle States

State	Control objective	Evidence
Created	Capture provenance, owner, purpose and classification.	Ingestion ledger and catalog entry.
Active	Meet product SLO and approved access purpose.	Usage, quality and access logs.
Inactive	Reduce compute and access while preserving required data.	Lifecycle state and consumer analysis.
Archived	Retain in lower-access storage with tested restoration.	Inventory, retention and restore test.
Held	Prevent deletion while legal or regulatory hold is active.	Case ID, approver and hold audit.
Eligible for deletion	All obligations expired and dependencies resolved.	Disposition approval and impact analysis.
Deleted or anonymized	Data and derived copies disposed according to policy.	Deletion certificate and exception list.

8.2 Retention Architecture

Retention schedules are stored as governed metadata and translated into bucket lifecycle rules, object retention settings, table and partition expiration or workflow-driven deletion. Irreversible bucket retention lock is used only after legal approval, testing and confirmation that the period cannot require reduction. Product owners cannot override enterprise record schedules through ad hoc table settings.

The raw layer often has the longest replay-oriented retention, but this is not universal. Sensitive personal data may have a shorter lawful period than aggregated products. Derived tables inherit purpose and deletion obligations unless they are demonstrably anonymized. The catalog records the schedule ID and deletion method for every certified product.

8.3 Archival and Restore

Control	Standard
Eligibility	No active SLO-dependent consumer and retention still applies.
Format	Open, documented and independently verifiable.
Index	Dataset, partition, date range, schema version, checksum and owner.
Restore objective	Defined by service tier and tested at least annually for critical archives.
Security	Same or stronger classification than source; privileged restore identity.
Cost	Storage class selected from measured restore frequency and retrieval economics.

8.4 Legal Hold

A legal hold suspends normal deletion for identified records without granting broader access. The legal function supplies case identifier, scope, custodian, start date and release authority. Cloud Storage object holds and retention policies can support

immutable records. BigQuery records may require protected snapshot/export or a dedicated held dataset depending on scope and legal evidence requirements. Hold application and release are audited and separated from ordinary data engineering roles.

8.5 Privacy Deletion

Receive a verified request and determine lawful scope, identity keys and jurisdictions.

Resolve the data subject across source-aligned and product identifiers.

Identify affected assets through lineage, product metadata and searchable indexes.

Apply source-system deletion or suppression according to authority.

Delete, anonymize or cryptographically render inaccessible eligible platform copies.

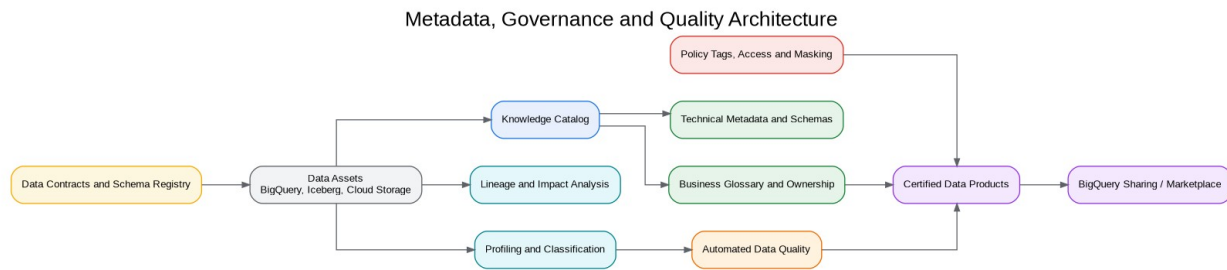
Propagate tombstones to downstream products and prevent replay from resurrecting deleted data.

Record exceptions for legal hold or statutory retention with owner and expiry.

Issue a deletion certificate containing scope, systems, timestamp, outcome and exceptions.

9. Metadata, Knowledge Catalog and Governance Architecture

Metadata is part of the product interface. Technical discovery without ownership and meaning is insufficient; business definitions without executable lineage and quality are equally insufficient. The governance plane combines automatic harvesting with explicit domain accountability.



Metadata, governance and quality architecture

Figure: Metadata, governance and quality architecture

9.1 Metadata Architecture

Metadata class	Examples	System of record
Technical	Projects, datasets, tables, columns, schemas, formats and partitions	Cloud services and Knowledge Catalog inventory
Operational	Runs, freshness, volume, failures, lag and cost	Monitoring, logging and product telemetry
Business	Definitions, owner, steward, domain, purpose and KPI meaning	Knowledge Catalog glossary and product repository
Security	Classification, policy tags, masking policy, residency and legal basis	Security metadata and catalog
Quality	Rules, scores, exceptions, certification and remediation	Quality service and product scorecard
Lineage	Source, transformation, destination, field mapping and consumer dependency	Knowledge Catalog lineage plus custom lineage events
Contract	Schema, compatibility, SLO, interface, version and deprecation	Version-controlled contract repository
Lifecycle	Retention schedule, archive class, hold and deletion method	Records system and catalog

9.2 Knowledge Catalog

Knowledge Catalog is the enterprise discovery and governance surface. Automatic inventory provides broad technical coverage, while domains add business context, ownership, classifications, glossary relationships and product status. Search permissions must not reveal sensitive metadata beyond policy. Catalog completeness is measured, not assumed.

Mandatory metadata for production products includes domain, product name, owner, steward, support channel, description, grain, refresh target, quality status, classification, residency, retention schedule, source lineage, contract location, version, lifecycle status and approved access path. Assets missing mandatory fields cannot be certified.

9.3 Dataplex Capabilities

Dataplex capabilities and APIs underpin inventory, profiling, quality and lineage functions that are now presented under Knowledge Catalog. The architecture uses automatic data quality for new implementations rather than legacy data-quality tasks. Data-quality rules and scans are managed as code where possible, and scan results feed product scorecards and publication gates.

9.4 Technical and Business Metadata

Area	Mandatory fields	Owner
Dataset	Domain, environment, location, classification,	Platform plus domain

Area	Mandatory fields	Owner
Table	retention and support tier Description, grain, keys, partition, owner, source and lifecycle	Domain data owner
Column	Business meaning, type, sensitivity, null semantics and allowed values	Data steward
Pipeline	Source, destination, schedule, SLO, runbook and code version	Technical owner
Product	Consumer, contract, SLO, quality, access, version and deprecation	Product owner
Metric	Definition, formula, aggregation, dimensions and approval	Business metric owner

9.5 Business Glossary

The glossary defines enterprise terms independently of individual tables. A term has a definition, domain, steward, synonyms, exclusions, examples and approved relationships. Terms can be linked to data products, columns and metrics. The governance council resolves cross-domain conflicts; domains retain authority for domain-specific terms. A glossary is not complete when it is merely imported from spreadsheets without stewardship and usage.

9.6 Data Lineage

Lineage supports impact analysis, audit, privacy deletion and incident response. Native lineage is used where supported. Custom pipelines emit lineage events that identify source and target assets, operation, code version, run ID and column mappings for critical products. Lineage completeness is an SLO for certified products. A broken lineage graph is treated as a production governance defect, not documentation debt.

9.7 Data Profiling

Profiling establishes distributions, null rates, uniqueness, common values, bounds and potential sensitive content before rule design. Profiles are sampled or full according to risk and cost. They are refreshed when source schema, population or product grain changes. Profile results must not expose sensitive sample values to unauthorized users.

9.8 Data-Quality Architecture

Quality rules execute at source receipt, layer transition and product publication. Transport integrity and schema validity are ingestion controls. Domain correctness, referential integrity and business reconciliation belong in silver and gold. Certified products add freshness, completeness, validity, consistency, uniqueness and accuracy objectives. Critical failures block publication; noncritical failures lower score and create a steward workflow.

Dimension	Example rule	Tier 0 threshold	Failure action
Completeness	Required customer ID is populated	100%	Block publication and alert owner.
Validity	Country code exists in approved reference	99.99%	Quarantine invalid records and investigate.
Uniqueness	Order ID unique within source domain	100%	Block merge and reconcile source.
Consistency	Order total equals line sum plus tax	99.99%	Block certified product.
Timeliness	Latest source window available	99.9% within SLO	Declare freshness incident.
Referential integrity	Account references existing customer	99.99%	Quarantine or use approved unknown member.
Accuracy	Finance debit equals credit control	100%	Block regulatory publication.

9.9 Quality Score and Certification

Score band	Status	Consumer behavior
99.5-100	Certified	Approved for executive, regulatory or operational use within stated scope.

Score band	Status	Consumer behavior
98-99.49	Governed	Usable with documented exceptions; not for highest-criticality decisions.
95-97.99	Conditional	Consumer warning and remediation plan required.
Below 95	Uncertified	Not published as an enterprise product.

10. Data Contracts, Schema Registry, Ownership and Certification

A data contract is a versioned agreement between producer and consumers. It covers more than schema: ownership, semantics, keys, update behavior, quality, latency, security, retention, compatibility and deprecation. Contracts are stored in source control and validated in CI and at runtime.

10.1 Data Contract Contents

Section	Required content
Identity	Domain, product/feed name, owner, steward, version and status.
Interface	Topic, table, view, API or file path and supported access method.
Schema	Fields, types, nullability, keys, grain, allowed values and classification.
Semantics	Business meaning, units, time zone, calculation and source authority.
Change	Compatibility mode, notice period, migration path and deprecation.
SLO	Freshness, availability, quality, support hours and recovery tier.
Security	Purpose, eligible groups, row/column policy, masking and residency.
Lifecycle	Retention, archive, deletion and legal-hold behavior.
Operations	Monitoring, runbook, escalation and consumer notification.
Cost	Cost owner, unit metric and consumer cost considerations.

10.2 Schema Registry

Event schemas use an approved registry and compatibility mode. Pub/Sub schemas can validate supported event representations. Batch and table schemas are stored in the contract repository and compared during CI and ingestion. The registry must not become a duplicate business glossary; it controls machine compatibility, while Knowledge Catalog provides broader context and discovery.

10.3 Data Ownership and Stewardship

Role	Accountability
Business data owner	Purpose, lawful use, product priority, access approval and risk acceptance.
Data-product owner	Roadmap, consumer contract, SLO, cost and lifecycle.
Data steward	Definitions, quality rules, reference data and issue remediation.
Technical owner	Pipeline, table, deployment, monitoring and recovery.
Platform owner	Shared services, templates, reliability and platform cost.
Security owner	Control design, threat assessment, exceptions and incident response.
Privacy/records owner	Legal basis, retention, hold and deletion policy.
Consumer owner	Approved purpose, downstream controls and usage compliance.

10.4 Data Certification

Product canvas approved with owner, consumers and business value.

Contract versioned and schema compatibility tested.

Classification, residency, retention and legal basis recorded.

End-to-end lineage visible and critical column mappings verified.

Quality rules pass representative and peak-volume data.

Access interfaces, row/column controls and masking tested with positive and negative identities.

Freshness, availability, recovery, monitoring and runbooks pass production readiness review.

Cost baseline, owner and budget threshold approved.

Catalog entry marked certified and publication channel enabled.

Quarterly recertification and event-driven review scheduled.



Data-product lifecycle and governance gates

Figure: Data-product lifecycle and governance gates

11. Data Products, Marketplace and Enterprise Access

A data product is a supported interface with a consumer promise. Tables may be implementation details. A product can expose one or more authorized views, linked datasets, streams, APIs or semantic models. The interface is stable, versioned, documented and observable.

11.1 Data Product Types

Type	Example	Primary interface
Source-aligned	ERP order change feed	Bronze table or event stream for approved engineering consumers.
Domain entity	Customer master profile	Silver table and authorized views.
Analytical product	Daily sales performance	Gold/curated tables and semantic model.
Regulatory product	Capital or compliance report dataset	Certified dataset with immutable publication evidence.
Operational product	Fraud feature lookup	Bigtable, Spanner or protected API.
AI-ready product	Grounded knowledge corpus	Curated text/chunks, embeddings metadata and access policy.
Reference product	Country, currency or product hierarchy	Versioned table with steward ownership.

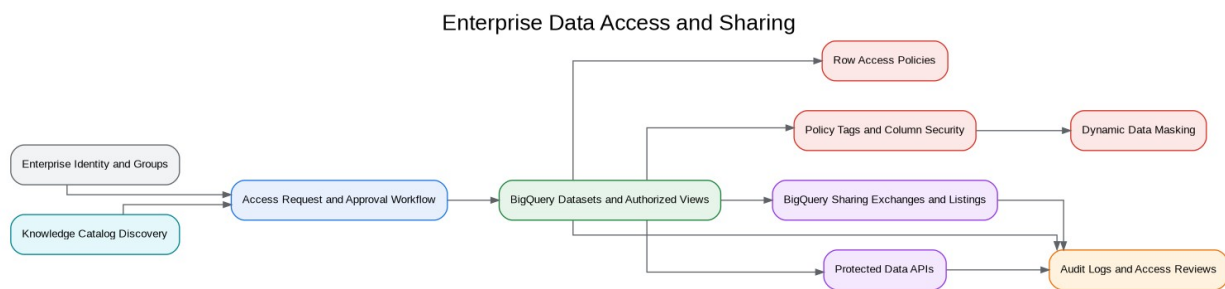
11.2 Data Marketplace

The marketplace combines discovery in Knowledge Catalog with governed publication through BigQuery sharing and approved APIs. Marketplace entries show purpose, owner, sample schema, classification, quality, SLO, cost model, access conditions and consumer responsibilities. Search does not imply access. Request workflows capture purpose and duration and route to owner, security or privacy approval based on classification.

11.3 BigQuery Sharing

Exchanges group products by domain or audience. Listings expose a stable dataset or stream contract. Subscribers receive linked datasets where supported, reducing unmanaged copies. Publishers remain responsible for product SLO and schema compatibility; subscribers remain responsible for approved use and downstream access. Location alignment and feature support are validated before publication.

11.4 Enterprise-Data Access Model



Enterprise data access and sharing model

Figure: Enterprise data access and sharing model

Access path	Use	Control
Dataset IAM	Small number of producer and operator groups	Avoid broad consumer access to base datasets.
Authorized view or dataset	Stable filtered or joined consumer interface	Publisher controls source access and interface.
Row access policy	Restrict records by region, legal entity, tenant or entitlement	Use group/attribute logic and test policy combinations.
Column-level security	Restrict raw sensitive columns	Policy tags and fine-grained reader roles.

Access path	Use	Control
Dynamic masking	Expose transformed values by role	Data policies, masking rules and policy tags.
BigQuery sharing	Cross-project or cross-organization product publication	
Protected API	Operational or externally controlled access	Apigee/Cloud Run, authentication, quotas and audit.
Export	Only when consumer cannot use managed access	Time-bound, encrypted, contract-controlled and monitored.

12. Data Classification, Fine-Grained Security and Encryption

Data controls are layered. Project and perimeter boundaries reduce blast radius. Dataset IAM controls broad roles. Authorized views provide stable interfaces. Row access policies filter records. Policy tags and column-level access restrict sensitive columns. Dynamic masking transforms visible values according to role. Encryption protects storage and approved key-control requirements. Audit and periodic review prove that the design remains effective.

12.1 Data Classification

Class	Examples	Minimum controls
Public	Approved public reference data	Integrity, provenance and controlled publication.
Internal	Operational metadata and nonpublic reference data	Employee groups, audit and standard encryption.
Confidential	Customer, commercial and employee records	Need-to-know groups, private access, DLP classification and monitoring.
Restricted	Payment, health, government ID and sensitive HR	Dedicated projects/perimeters, policy tags, masking, CMEK where required and enhanced review.
Regulated	Records governed by specific retention or reporting law	All restricted controls plus evidence, hold and regulatory approval.

12.2 Row-Level Security

Row policies are used when the same table serves multiple approved populations and the filter is stable, testable and performant. Entitlement logic is maintained through managed groups or reference tables rather than hard-coded individuals. Policies are version-controlled and tested for policy intersection, privileged access and service-account behavior. Highly complex consumer-specific logic may be safer in authorized views.

12.3 Column-Level Security and Policy Tags

Policy tags classify columns and control access to original values. The taxonomy aligns to enterprise classification and data categories such as direct identifiers, payment data, health data and secrets. Tag assignment is deployed with schema code and validated in CI. Access to unmasked values is narrowly granted to governed groups.

12.4 Dynamic Data Masking

Dynamic masking provides role-dependent representations such as null, default, last-four, hash or email masking. It is not anonymization and does not change the stored base value. Masking policies are selected based on consumer purpose and reidentification risk. Negative tests confirm that users without privileged roles never receive raw values through views, extracts or indirect query paths.

12.5 Authorized Views

Authorized views are the preferred interface when consumers need a stable subset, calculated field or cross-table join without base-table access. Views are owned by the publishing domain, tested for performance and protected from accidental replacement. Breaking changes follow product versioning and deprecation.

12.6 Encryption

Layer	Default	Enhanced control
Cloud Storage	Google-managed encryption	CMEK for restricted/regulatory buckets; key rotation and separation of duties.
BigQuery	Google-managed encryption	CMEK at dataset/table-supported boundary where policy requires.
BigLake/Iceberg	Storage encryption plus service access controls	CMEK and external-engine key/access validation.
Pub/Sub	Service encryption	CMEK for approved topics where supported and required.
Dataflow	Service encryption and encrypted disks	CMEK and no public worker IPs where required.

Layer	Default	Enhanced control
Operational databases	Managed encryption	CMEK and database-specific backup key strategy.
Transfer	TLS or private connectivity	Partner encryption, signed manifests and customer-controlled keys where contract requires.

13. Data-Platform SLOs, Monitoring and Cost Model

Platform reliability is expressed through service-level indicators that consumers can understand: freshness, availability, completeness, quality, access latency and recovery. Infrastructure metrics are necessary but not sufficient. Each certified product publishes an SLO and an error budget with a named owner.

13.1 SLO Catalogue

SLI	Measurement	Tier 0 target	Tier 1 target	Tier 2 target
Freshness	Current source watermark minus product watermark	99.9% within 5 minutes	99.5% within 30 minutes	99% within 24 hours
Pipeline success	Successful scheduled windows / expected windows	99.95%	99.9%	99%
Availability	Successful governed queries or API requests	99.99%	99.9%	99.5%
Completeness	Accepted source records or control total	100% for financial controls	99.99%	99.5%
Quality	Weighted mandatory rule pass rate	99.9%	99.5%	98%
Discoverability	Mandatory metadata fields complete	100%	100%	95%
Access fulfillment	Approved request to usable access	4 business hours	1 business day	3 business days
Recovery	Restore/replay within RTO/RPO	<1h / <5m	<4h / <15m	<24h / <24h

13.2 Monitoring Architecture

Layer	Signals	Alert examples
Transfer	Objects, bytes, checksum, duration and failures	Missing batch, partial inventory, repeated transfer error.
Streaming	Publish rate, backlog, oldest unacked, watermark and DLQ	Backlog age exceeds freshness budget; DLQ spike.
CDC	Source lag, stream state, schema change and merge delay	Lag exceeds threshold; log retention risk.
Storage	Growth, object count, lifecycle, retention and egress	Unexpected region egress; lifecycle not reducing old data.
BigQuery	Jobs, slots, queue, bytes, storage and failures	Queued production workload; anomalous scan cost.
Quality	Rule pass, failed rows, score and exceptions	Critical rule blocks publication.
Metadata	Coverage, stale ownership and lineage gaps	Certified product missing lineage or owner.
Security	Denied access, policy change and data exfiltration indicators	Unexpected raw-column access or perimeter violation.
Cost	Daily spend, unit cost, forecast and anomaly	Product exceeds budget or unit cost trend.

13.3 Product Scorecard

Category	Measure	Weight
Reliability	Freshness and availability SLO	25%
Quality	Mandatory and advisory rule score	25%
Governance	Metadata, lineage and ownership completeness	15%

Category	Measure	Weight
Security	Access review and policy conformance	15%
Consumer	Usage, satisfaction and incidents	10%
FinOps	Budget and unit-cost trend	10%

13.4 Cost Model

The architecture does not embed current price values. Cost is calculated from measurable workload variables and current Google Cloud pricing. Each product has an expected monthly range, a unit metric and optimization owner.

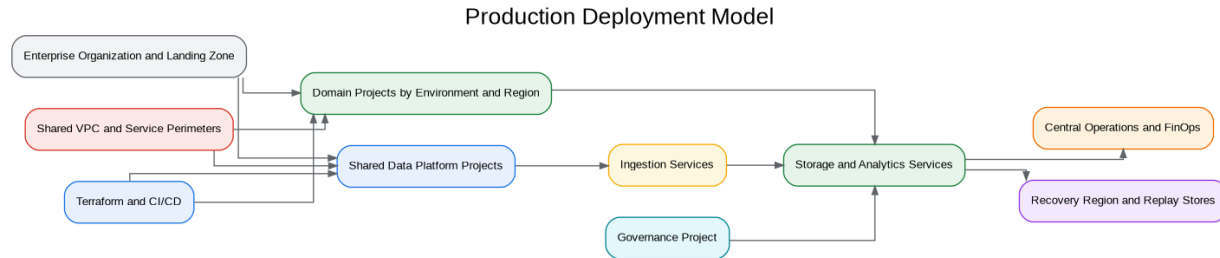
Cost component	Model variables	Primary controls
Cloud Storage	GB-month by class, operations, retrieval and egress	Lifecycle, compression, locality, compaction and retention.
BigQuery storage	Active and long-term GB-month	Table lifecycle, duplicate copy elimination and snapshots.
BigQuery compute	On-demand bytes processed or capacity/slot usage	Partitioning, clustering, reservations, workload assignment and query review.
Streaming	Pub/Sub throughput/retention and Dataflow workers/runtime	Message size, retention, autoscaling, state and at-least-once choice.
CDC	Datastream processed volume plus target merge compute	Filter unused objects and tune merge cadence.
Governance	Metadata storage/scans and profile/quality execution	Scope scans by risk and reuse rules.
Network	Inter-region, internet and cross-cloud egress	Regional locality and transfer architecture.
Logging	Log ingestion and retention	Exclusions, sampling and tiered retention without losing audit evidence.
Operational stores	Node/compute, storage, backups and network	Right-size, autoscale, workload-specific retention and capacity review.

13.5 Unit Economics

Unit metric	Formula
Batch ingestion cost per TB	Monthly batch transfer + processing + load cost / accepted TB.
Streaming cost per million events	Pub/Sub + Dataflow + serving writes / accepted million events.
Data product cost per active consumer	Attributed storage + compute + operations / monthly active consumers.
BigQuery cost per successful query	Attributed query compute / successful governed queries.
Quality cost per billion rows	Profile and rule execution cost / billion assessed rows.
Archive cost per retained TB-year	Storage + inventory + restore testing / retained TB-year.

14. Production Deployment Model

The platform inherits organization, network, identity and security foundations from Volume 02. Shared platform projects host reusable control-plane capabilities; domain projects host data-plane assets and product workloads. Production and nonproduction are separated. Regional data planes align to residency and recovery. Infrastructure, schemas, contracts and policies are promoted through CI/CD.



Production deployment model

Figure: Production deployment model

14.1 Project Model

Project type	Example purpose	Ownership
Data platform control	Catalog integration, shared metadata, policy automation and product registry	Central platform
Ingestion shared service	Reusable transfer, validation, templates and DLQ services	Central platform
Domain data project	BigQuery datasets, buckets, topics and product pipelines for one domain/environment/region	Domain team within platform guardrails
Restricted domain project	Restricted data assets inside stronger service perimeter	Domain plus security
Operations project	Monitoring workspaces, dashboards and alert routing	SRE/platform operations
KMS project	Regional keys and separation of key administrators	Security platform
Sharing project	Exchanges, listings and controlled publication	Marketplace/platform team
Sandbox project	Time-bound exploration with quotas and no restricted raw data	Managed self-service

14.2 Environment Strategy

Environment	Data	Purpose	Promotion gate
Development	Synthetic or masked samples	Code, schema and contract development	Unit, lint, policy and schema tests.
Integration	Representative nonproduction data	Cross-service integration and migration tests	End-to-end reconciliation and security tests.
Performance	Generated or approved representative scale	Peak, skew, cost and failure testing	Capacity and cost acceptance.
Staging	Production-like configuration and masked/sampled data	Release rehearsal and operational readiness	Change approval and runbook validation.
Production	Approved business data	Consumer service	Automated deployment, monitoring and rollback.
Sandbox	Nonrestricted, time-bound copies	Exploration	Expiry and no production dependency.

14.3 Regional and Recovery Model

Each data domain has a home geography and approved primary region. Data does not cross a residency boundary solely for centralized convenience. Critical products define a recovery region and a service-specific strategy: replay from retained events, rebuild from raw objects, copy or snapshot protected BigQuery tables, and reconstruct infrastructure from Terraform. Global metadata may describe assets across regions, but search and metadata visibility remain access-controlled.

14.4 Deployment Sequence

Create domain and shared projects through the landing-zone project factory.

Attach Shared VPC, service perimeter, logging, monitoring, budgets and KMS according to classification.

Deploy buckets, datasets, topics, service accounts and base IAM through approved modules.

Deploy contract, schema, policy tags, row policies and masking policies.

Deploy ingestion and transformation artifacts with immutable version identifiers.

Run automated schema, security, quality, reconciliation and performance tests.

Execute production-readiness review and approve SLO, runbook, cost owner and recovery plan.

Promote release, validate telemetry and record deployment evidence.

Publish or update the data product in Knowledge Catalog and BigQuery sharing.

Monitor error budget and rollback or disable publication on critical failure.

14.5 Production Acceptance Checklist

Control domain	Acceptance evidence
Ownership	Business owner, product owner, steward and technical owner approved.
Architecture	Approved service selection and ADR references.
Security	Classification, IAM, perimeter, masking, encryption and threat test passed.
Data	Contract, schema evolution, reconciliation and quality rules passed.
Operations	SLO, dashboards, alerts, runbook, on-call and support path active.
Recovery	Replay, restore or rebuild demonstrated against target RTO/RPO.
FinOps	Budget, labels, forecast and unit-cost baseline recorded.
Governance	Catalog, glossary, lineage, retention and certification complete.
Change	CI/CD, rollback and consumer notification tested.

15. Architecture Decision Records

The following decisions form the enterprise baseline. Each implementation references the applicable ADRs and records any exception, compensating control and review date.

ADR-DP-001: BigQuery is the primary analytical warehouse

Status: Approved reference baseline

Architecture domain: Enterprise data platform

Problem. The enterprise needs petabyte analytical scale, fine-grained governance and managed operations.

Alternatives considered. BigQuery native tables; Iceberg-only lakehouse; third-party warehouse.

Decision. Use BigQuery native tables for the majority of silver, gold, curated and semantic-serving data.

Rationale and benefits. Native management and security reduce operating complexity. Iceberg is retained for real interoperability needs.

Trade-offs and risks. Cost can grow with poor query design; regional boundaries and quotas apply.

Mitigations. Partition/cluster standards, workload isolation, budgets and service-limit validation.

Implementation implications. Reference modules, contracts, monitoring and validation must implement this decision.

Review triggers. Material service change, regulatory change, repeated SLO breach, cost anomaly or approved exception pattern.

Review date. Quarterly architecture review.

ADR-DP-002: Cloud Storage is the immutable raw and archive foundation

Status: Approved reference baseline

Architecture domain: Enterprise data platform

Problem. Source evidence and open objects must be retained independently of compute.

Alternatives considered. Cloud Storage; BigQuery-only; self-managed object store.

Decision. Use Cloud Storage for landing, quarantine, raw, replay and archive.

Rationale and benefits. Durable object semantics, open formats and lifecycle controls.

Trade-offs and risks. Object management and small-file overhead require discipline.

Mitigations. Compaction, manifest, inventory, lifecycle and access standards.

Implementation implications. Reference modules, contracts, monitoring and validation must implement this decision.

Review triggers. Material service change, regulatory change, repeated SLO breach, cost anomaly or approved exception pattern.

Review date. Quarterly architecture review.

ADR-DP-003: Iceberg is selective, not universal

Status: Approved reference baseline

Architecture domain: Enterprise data platform

Problem. Multiple engines sometimes need one authoritative open table.

Alternatives considered. BigQuery native; Apache Iceberg; duplicate copies per engine.

Decision. Use Apache Iceberg managed/external tables only when multi-engine or customer-owned storage is required.

Rationale and benefits. Preserves interoperability without imposing lakehouse complexity everywhere.

Trade-offs and risks. Writer compatibility and metadata operations add risk.

Mitigations. Approved writers, one catalog authority, compaction and compatibility testing.

Implementation implications. Reference modules, contracts, monitoring and validation must implement this decision.

Review triggers. Material service change, regulatory change, repeated SLO breach, cost anomaly or approved exception pattern.

Review date. Quarterly architecture review.

ADR-DP-004: Layer names have explicit contracts

Status: Approved reference baseline

Architecture domain: Enterprise data platform

Problem. Raw, bronze, silver, gold and curated are often used inconsistently.

Alternatives considered. Team-defined layers; enterprise definitions; no layers.

Decision. Adopt the layer definitions in this volume.

Rationale and benefits. Reduces duplicate zones and clarifies ownership and exit criteria.

Trade-offs and risks. Migration from existing terminology requires mapping.

Mitigations. Maintain a crosswalk and certify by controls, not folder name.

Implementation implications. Reference modules, contracts, monitoring and validation must implement this decision.

Review triggers. Material service change, regulatory change, repeated SLO breach, cost anomaly or approved exception pattern.

Review date. Quarterly architecture review.

ADR-DP-005: Batch is the default latency pattern

Status: Approved reference baseline

Architecture domain: Enterprise data platform

Problem. Streaming adds state, replay and cost complexity.

Alternatives considered. Batch by default; streaming by default; case-by-case with no standard.

Decision. Use the slowest pattern that meets the business decision latency.

Rationale and benefits. Improves reliability and unit economics.

Trade-offs and risks. Some teams may understate latency need.

Mitigations. Business owner signs latency class and reviews consumer outcomes.

Implementation implications. Reference modules, contracts, monitoring and validation must implement this decision.

Review triggers. Material service change, regulatory change, repeated SLO breach, cost anomaly or approved exception pattern.

Review date. Quarterly architecture review.

ADR-DP-006: Pub/Sub is the cloud-native event backbone

Status: Approved reference baseline

Architecture domain: Enterprise data platform

Problem. Cloud consumers need decoupled event delivery.

Alternatives considered. Pub/Sub; managed Kafka; direct producer-to-consumer calls.

Decision. Use Pub/Sub for new Google Cloud-native events; bridge existing Kafka where needed.

Rationale and benefits. Managed scale and independent subscriptions.

Trade-offs and risks. At-least-once delivery requires idempotent consumers.

Mitigations. Stable event IDs, schemas, DLQ and replay archive.

Implementation implications. Reference modules, contracts, monitoring and validation must implement this decision.

Review triggers. Material service change, regulatory change, repeated SLO breach, cost anomaly or approved exception pattern.

Review date. Quarterly architecture review.

ADR-DP-007: Dataflow handles complex event processing

Status: Approved reference baseline

Architecture domain: Enterprise data platform

Problem. Stateful streaming and large parallel batch require a managed engine.

Alternatives considered. Dataflow; Spark; custom services.

Decision. Use Dataflow when Beam semantics, windows, state or scale are required.

Rationale and benefits. Managed autoscaling and common model.

Trade-offs and risks. Engineering complexity for simple jobs.

Mitigations. Use SQL or Cloud Run for simpler transformations.

Implementation implications. Reference modules, contracts, monitoring and validation must implement this decision.

Review triggers. Material service change, regulatory change, repeated SLO breach, cost anomaly or approved exception pattern.

Review date. Quarterly architecture review.

ADR-DP-008: Datastream is preferred for supported CDC

Status: Approved reference baseline

Architecture domain: Enterprise data platform

Problem. Custom CDC platforms create connector and recovery burden.

Alternatives considered. Datastream; Debezium/Kafka; batch extracts; third-party CDC.

Decision. Use Datastream where source, target and required behavior are supported.

Rationale and benefits. Reduces platform operations.

Trade-offs and risks. Coverage and schema behavior require validation.

Mitigations. Pilot, reconcile and retain an approved fallback.

Implementation implications. Reference modules, contracts, monitoring and validation must implement this decision.

Review triggers. Material service change, regulatory change, repeated SLO breach, cost anomaly or approved exception pattern.

Review date. Quarterly architecture review.

ADR-DP-009: Domain projects own production data products

Status: Approved reference baseline

Architecture domain: Enterprise data platform

Problem. One central project creates access, quota and cost blast radius.

Alternatives considered. Central project; project per table; domain/environment projects.

Decision. Use domain, environment, geography and sensitivity as project boundaries.

Rationale and benefits. Balances autonomy, governance and attribution.

Trade-offs and risks. More projects require automation.

Mitigations. Project factory and standard modules.

Implementation implications. Reference modules, contracts, monitoring and validation must implement this decision.

Review triggers. Material service change, regulatory change, repeated SLO breach, cost anomaly or approved exception pattern.

Review date. Quarterly architecture review.

ADR-DP-010: Knowledge Catalog is the enterprise governance surface

Status: Approved reference baseline

Architecture domain: Enterprise data platform

Problem. Metadata is fragmented across tools and teams.

Alternatives considered. Native catalog; third-party catalog; spreadsheets.

Decision. Use Knowledge Catalog as the Google Cloud governance surface and integrate external enterprise metadata where required.

Rationale and benefits. Native inventory, lineage, glossary, profiling and quality integration.

Trade-offs and risks. Coverage and product evolution must be managed.

Mitigations. Completeness SLO, integration architecture and quarterly feature review.

Implementation implications. Reference modules, contracts, monitoring and validation must implement this decision.

Review triggers. Material service change, regulatory change, repeated SLO breach, cost anomaly or approved exception pattern.

Review date. Quarterly architecture review.

ADR-DP-011: Automatic data quality is the default managed quality service

Status: Approved reference baseline

Architecture domain: Enterprise data platform

Problem. Quality must be repeatable and visible.

Alternatives considered. Knowledge Catalog automatic quality; custom SQL only; legacy tasks; third-party.

Decision. Use automatic data quality and reusable rules, with custom SQL where needed.

Rationale and benefits. Managed execution and score integration.

Trade-offs and risks. Complex business rules may need custom processing.

Mitigations. Separate transport, domain and product quality layers.

Implementation implications. Reference modules, contracts, monitoring and validation must implement this decision.

Review triggers. Material service change, regulatory change, repeated SLO breach, cost anomaly or approved exception pattern.

Review date. Quarterly architecture review.

ADR-DP-012: Contracts are stored and validated as code

Status: Approved reference baseline

Architecture domain: Enterprise data platform

Problem. Schema and semantic changes cause consumer incidents.

Alternatives considered. Wiki documents; schema only; versioned contracts.

Decision. Use version-controlled contracts with CI and runtime validation.

Rationale and benefits. Creates enforceable producer-consumer expectations.

Trade-offs and risks. Requires governance and migration discipline.

Mitigations. Templates, compatibility tests and deprecation policy.

Implementation implications. Reference modules, contracts, monitoring and validation must implement this decision.

Review triggers. Material service change, regulatory change, repeated SLO breach, cost anomaly or approved exception pattern.

Review date. Quarterly architecture review.

ADR-DP-013: Authorized interfaces replace broad base-table access

Status: Approved reference baseline

Architecture domain: Enterprise data platform

Problem. Direct table grants expose implementation detail and sensitive data.

Alternatives considered. Base dataset access; views; extracts.

Decision. Publish authorized views/datasets, row policies, masked columns or APIs.

Rationale and benefits. Supports least privilege and stable contracts.

Trade-offs and risks. View complexity can affect performance.

Mitigations. Performance tests, materialization and interface versioning.

Implementation implications. Reference modules, contracts, monitoring and validation must implement this decision.

Review triggers. Material service change, regulatory change, repeated SLO breach, cost anomaly or approved exception pattern.

Review date. Quarterly architecture review.

ADR-DP-014: BigQuery sharing is the default governed marketplace mechanism

Status: Approved reference baseline

Architecture domain: Enterprise data platform

Problem. Consumers need cross-project and cross-organization access without unmanaged copies.

Alternatives considered. Direct IAM; exports; BigQuery sharing; API.

Decision. Use exchanges and listings for supported analytical datasets and streams.

Rationale and benefits. Linked access and publisher control.

Trade-offs and risks. Location and feature constraints apply.

Mitigations. Validate consumer, location, contract and subscriber governance.

Implementation implications. Reference modules, contracts, monitoring and validation must implement this decision.

Review triggers. Material service change, regulatory change, repeated SLO breach, cost anomaly or approved exception pattern.

Review date. Quarterly architecture review.

ADR-DP-015: CMEK is risk-based

Status: Approved reference baseline

Architecture domain: Enterprise data platform

Problem. Universal CMEK increases key dependencies without equal value.

Alternatives considered. Google-managed keys; universal CMEK; selective CMEK.

Decision. Use CMEK for restricted/regulatory workloads or explicit customer control requirements.

Rationale and benefits. Aligns key-control burden to risk.

Trade-offs and risks. Classification errors could omit required CMEK.

Mitigations. Policy automation, inventory and security review.

Implementation implications. Reference modules, contracts, monitoring and validation must implement this decision.

Review triggers. Material service change, regulatory change, repeated SLO breach, cost anomaly or approved exception pattern.

Review date. Quarterly architecture review.

ADR-DP-016: Bigtable is conditional operational serving

Status: Approved reference baseline

Architecture domain: Enterprise data platform

Problem. Some products need high-throughput low-latency key access.

Alternatives considered. Bigtable; BigQuery; Spanner; cache.

Decision. Use Bigtable only for access-pattern-driven wide-column/key-value workloads.

Rationale and benefits. Scales serving independently of analytical compute.

Trade-offs and risks. Schema mistakes create hotspots and weak query flexibility.

Mitigations. Row-key review, load test and app-profile monitoring.

Implementation implications. Reference modules, contracts, monitoring and validation must implement this decision.

Review triggers. Material service change, regulatory change, repeated SLO breach, cost anomaly or approved exception pattern.

Review date. Quarterly architecture review.

ADR-DP-017: Spanner is conditional for globally consistent transactions

Status: Approved reference baseline

Architecture domain: Enterprise data platform

Problem. Global applications may need relational consistency and scale.

Alternatives considered. Spanner; regional PostgreSQL; sharded databases.

Decision. Use Spanner when consistency, availability and scale justify its model.

Rationale and benefits. Mission-critical relational capability at scale.

Trade-offs and risks. Higher design and cost complexity.

Mitigations. Workload proof, schema review and configuration testing.

Implementation implications. Reference modules, contracts, monitoring and validation must implement this decision.

Review triggers. Material service change, regulatory change, repeated SLO breach, cost anomaly or approved exception pattern.

Review date. Quarterly architecture review.

ADR-DP-018: AlloyDB is the preferred demanding PostgreSQL-compatible option

Status: Approved reference baseline

Architecture domain: Enterprise data platform

Problem. Some operational products require PostgreSQL compatibility.

Alternatives considered. AlloyDB; Cloud SQL; Spanner PostgreSQL dialect.

Decision. Use AlloyDB for demanding regional PostgreSQL-compatible workloads.

Rationale and benefits. Application compatibility and managed HA.

Trade-offs and risks. Not globally distributed like Spanner.

Mitigations. Connection, backup, read-pool and DR design.

Implementation implications. Reference modules, contracts, monitoring and validation must implement this decision.

Review triggers. Material service change, regulatory change, repeated SLO breach, cost anomaly or approved exception pattern.

Review date. Quarterly architecture review.

ADR-DP-019: Retention and deletion are metadata-driven

Status: Approved reference baseline

Architecture domain: Enterprise data platform

Problem. Manual lifecycle settings lead to noncompliance and excess cost.

Alternatives considered. Manual rules; platform defaults only; policy-driven lifecycle.

Decision. Map record schedules and privacy obligations to deployable lifecycle controls.

Rationale and benefits. Consistent evidence and automated disposition.

Trade-offs and risks. Incorrect mapping can cause premature deletion.

Mitigations. Legal approval, dry run, hold controls and deletion certificates.

Implementation implications. Reference modules, contracts, monitoring and validation must implement this decision.

Review triggers. Material service change, regulatory change, repeated SLO breach, cost anomaly or approved exception pattern.

Review date. Quarterly architecture review.

ADR-DP-020: Products are certified through measurable gates

Status: Approved reference baseline

Architecture domain: Enterprise data platform

Problem. A catalog label alone does not establish trust.

Alternatives considered. Informal publication; documentation review; automated certification gates.

Decision. Require contract, lineage, quality, security, SLO, recovery and cost evidence.

Rationale and benefits. Creates reliable enterprise interfaces.

Trade-offs and risks. Initial delivery takes longer.

Mitigations. Tiered certification and reusable automated evidence.

Implementation implications. Reference modules, contracts, monitoring and validation must implement this decision.

Review triggers. Material service change, regulatory change, repeated SLO breach, cost anomaly or approved exception pattern.

Review date. Quarterly architecture review.

Appendix A. Source-to-Target Mapping

The following representative mapping demonstrates the level of design detail required before implementation. Actual mappings are generated from the source registry and contract repository.

A.1 Oracle CRM CUSTOMER

Pattern. CDC
Ingestion. Datastream
Raw target. customer/raw/crm
Bronze target. customer_bronze.customer_change
Product target. customer_silver.customer
Business key. customer_id
Classification. Restricted
Freshness objective. <5 min

A.2 SAP S/4 SALES_ORDER

Pattern. Batch delta
Ingestion. Managed connector/file
Raw target. sales/raw/sap
Bronze target. sales_bronze.sales_order
Product target. sales_gold.daily_sales
Business key. sales_order_id
Classification. Confidential
Freshness objective. Hourly

A.3 Salesforce Account

Pattern. SaaS transfer
Ingestion. BQ Data Transfer or connector
Raw target. sales/raw/salesforce
Bronze target. sales_bronze.account
Product target. customer_silver.organization
Business key. account_id
Classification. Confidential
Freshness objective. Hourly

A.4 Kafka clickstream

Pattern. Streaming
Ingestion. Pub/Sub/Dataflow
Raw target. digital/replay/clickstream
Bronze target. digital_bronze.click_event
Product target. digital_gold.session_metrics
Business key. event_id
Classification. Internal
Freshness objective. <2 min

A.5 IoT telemetry

Pattern. Streaming
Ingestion. Pub/Sub/Dataflow
Raw target. operations/replay/iot
Bronze target. operations_bronze.telemetry
Product target. operations_gold.asset_health
Business key. device_id,event_ts
Classification. Confidential
Freshness objective. <1 min

A.6 Mainframe policy file

Pattern. Batch file
Ingestion. MFT/Cloud Storage/Dataflow
Raw target. policy/raw/mainframe
Bronze target. policy_bronze.policy
Product target. policy_silver.policy
Business key. policy_number
Classification. Restricted
Freshness objective. Daily

A.7 Workday worker

Pattern. SaaS API
Ingestion. Cloud Run/Workflows
Raw target. hr/raw/workday
Bronze target. hr_bronze.worker
Product target. hr_curated.workforce_summary
Business key. worker_id
Classification. Restricted
Freshness objective. Daily

A.8 Partner price file

Pattern. Partner file
Ingestion. Cloud Storage validation
Raw target. supply/raw/partner
Bronze target. supply_bronze.partner_price
Product target. supply_silver.supplier_price
Business key. partner_id,sku
Classification. Confidential
Freshness objective. Daily

A.9 AWS S3 web logs

Pattern. Cross-cloud transfer
Ingestion. Storage Transfer Service
Raw target. digital/raw/aws
Bronze target. digital_bronze.web_log
Product target. digital_gold.traffic
Business key. request_id
Classification. Internal
Freshness objective. Hourly

A.10 Payment events

Pattern. Streaming
Ingestion. Pub/Sub/Dataflow
Raw target. finance/replay/payment
Bronze target. finance_bronze.payment_event
Product target. finance_curated.payment_control
Business key. payment_id
Classification. Regulated
Freshness objective. <30 sec

A.11 Product reference API

Pattern. API

Ingestion. Cloud Run

Raw target. product/raw/reference

Bronze target. product_bronze.product

Product target. product_silver.product_master

Business key. product_id

Classification. Internal

Freshness objective. Hourly

A.12 Document repository

Pattern. File/object

Ingestion. Transfer + metadata extraction

Raw target. knowledge/raw/documents

Bronze target. knowledge_bronze.document

Product target. knowledge_curated.approved_corpus

Business key. document_id,version

Classification. Restricted

Freshness objective. Daily

Appendix B. Data-Layer Standards Matrix

B.1 Landing

Required metadata. source, batch, object generation, checksum

Required tests. completion, checksum, malware, manifest

Allowed consumers. ingestion service

Default lifecycle. delete after accepted or quarantined

B.2 Quarantine

Required metadata. error, rule, owner, correlation, remediation

Required tests. none before isolation

Allowed consumers. operations and steward

Default lifecycle. delete after resolution and evidence period

B.3 Raw

Required metadata. source, contract, schema, provenance, retention

Required tests. integrity and source reconciliation

Allowed consumers. restricted engineering/audit

Default lifecycle. retain per replay/legal requirement

B.4 Bronze

Required metadata. operation, source time, ingestion time, batch/offset

Required tests. type, key, duplicate, basic validity

Allowed consumers. domain engineering

Default lifecycle. retain for operational rebuild

B.5 Silver

Required metadata. domain entity, owner, history, references

Required tests. conformance, quality, referential integrity

Allowed consumers. approved domain consumers

Default lifecycle. domain retention

B.6 Gold

Required metadata. product, consumer, SLO, cost

Required tests. business rules, performance, freshness

Allowed consumers. named consumers

Default lifecycle. product retention

B.7 Curated

Required metadata. enterprise definition, certification, lineage

Required tests. strict quality, security and reconciliation

Allowed consumers. broad approved enterprise users

Default lifecycle. enterprise/regulatory schedule

B.8 Semantic

Required metadata. metric owner, formula, version

Required tests. metric test and reconciliation

Allowed consumers. BI and application consumers

Default lifecycle. version history

B.9 Sandbox

Required metadata. user, purpose, expiry, source reference

Required tests. sensitivity and expiry check

Allowed consumers. sandbox group

Default lifecycle. automatic expiry

B.10 Archive

Required metadata. schedule, checksum, schema, restore owner

Required tests. inventory and restore test

Allowed consumers. privileged restore

Default lifecycle. record schedule

Appendix C. BigQuery Design Standards

C.1 Project/dataset boundary

Standard. Separate by domain, environment, geography and materially different classification.

Anti-pattern. Avoid one enterprise dataset or dataset per table.

C.2 Naming

Standard. Use lower_snake_case for datasets/tables and stable business names.

Anti-pattern. Avoid vendor abbreviations in curated products.

C.3 Location

Standard. Set dataset location from residency and recovery architecture.

Anti-pattern. Do not mix locations in a product dependency without explicit design.

C.4 Partitioning

Standard. Use dominant business date/time predicate; require filters where appropriate.

Anti-pattern. Avoid date-sharded tables.

C.5 Clustering

Standard. Use proven selective filters/joins; most valuable column first.

Anti-pattern. Do not add four columns by default.

C.6 Table grain

Standard. Document one row meaning, keys and duplicate behavior.

Anti-pattern. No table without an explicit grain.

C.7 Nested fields

Standard. Use for bounded one-to-many structures queried together.

Anti-pattern. Avoid deeply nested structures that hide business entities.

C.8 Views

Standard. Use authorized views for stable consumer interfaces.

Anti-pattern. Do not expose base-table implementation as the product.

C.9 Materialized results

Standard. Use where repeated expensive logic and freshness allow.

Anti-pattern. Do not precompute without usage evidence.

C.10 Row security

Standard. Use stable entitlement dimensions and negative tests.

Anti-pattern. Avoid individual email logic.

C.11 Column security

Standard. Apply policy tags and least-privilege fine-grained roles.

Anti-pattern. Do not rely on naming conventions for sensitivity.

C.12 Masking

Standard. Use policy-driven dynamic masking for approved reduced views.

Anti-pattern. Masking is not anonymization.

C.13 Reservations

Standard. Separate critical BI/ELT/data science based on observed workload.

Anti-pattern. Do not purchase capacity before measuring.

C.14 Query standards

Standard. Explicit columns, partition filters, bounded joins and dry-run checks.

Anti-pattern. No production SELECT * in recurring jobs.

C.15 Schema deployment

Standard. DDL and policy changes are version-controlled and promoted through CI/CD.

Anti-pattern. No direct console-only production schema changes.

C.16 Recovery

Standard. Use time travel, snapshots, source/raw rebuild and tested runbooks according to tier.

Anti-pattern. Do not assume time travel alone meets all backup requirements.

Appendix D. Sample Dataset and Table Definitions

The package includes executable SQL files. The examples below illustrate the enterprise patterns and must be adapted for project IDs, regions, identities, policy-taxonomy IDs and retention schedules.

D.1 Dataset Definition

```
CREATE SCHEMA IF NOT EXISTS `acme-customer-prd.customer_silver`
OPTIONS (
  location = 'us-central1',
  description = 'Standardized customer domain entities',
  default_table_expiration_days = NULL,
  labels = [('domain','customer'),('environment','prd'),('classification','confidential')]
);
```

D.2 Partitioned and Clustered Table

```
CREATE TABLE IF NOT EXISTS `acme-customer-prd.customer_silver.customer_event` (
  event_id STRING NOT NULL,
  customer_id STRING NOT NULL,
  event_ts TIMESTAMP NOT NULL,
  event_date DATE NOT NULL,
  event_type STRING NOT NULL,
  region_code STRING,
  payload JSON,
  source_system STRING NOT NULL,
  schema_version STRING NOT NULL,
  ingestion_ts TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP()
)
PARTITION BY event_date
CLUSTER BY customer_id, event_type, region_code
OPTIONS (
  description = 'Standardized customer events; one row per source event',
  require_partition_filter = TRUE,
  labels = [('domain','customer'),('layer','silver'),('data_product','customer_events')]
);
```

D.3 Authorized View

```
CREATE OR REPLACE VIEW `acme-customer-prd.customer_access.customer_summary_v1` AS
SELECT
  customer_id,
  region_code,
  COUNTIF(event_type = 'PURCHASE') AS purchase_count_90d,
  MAX(event_ts) AS last_activity_ts
FROM `acme-customer-prd.customer_silver.customer_event`
WHERE event_date >= DATE_SUB(CURRENT_DATE(), INTERVAL 90 DAY)
GROUP BY customer_id, region_code;
```

D.4 Row Access Policy

```
CREATE OR REPLACE ROW ACCESS POLICY region_entitlement
ON `acme-customer-prd.customer_curated.customer_360`
GRANT TO ('group:customer-emea-readers@example.com')
FILTER USING (region_code = 'EMEA');
```

D.5 Snapshot Before High-Risk Change

```
CREATE SNAPSHOT TABLE `acme-customer-prd.customer_backup.customer_360_20260711`
CLONE `acme-customer-prd.customer_curated.customer_360`
OPTIONS (expiration_timestamp = TIMESTAMP_ADD(CURRENT_TIMESTAMP(), INTERVAL 30 DAY));
```

Appendix E. Validation Checklists

E.1 Source Onboarding

- ☐ Business and technical owners are named.
- ☐ Source authority and consumer use case are approved.
- ☐ Sensitivity, residency, legal basis and retention are recorded.
- ☐ Latency is justified by a business outcome.
- ☐ Volume, change rate and peak behavior are measured.
- ☐ Contract and compatibility mode are version-controlled.
- ☐ Reconciliation and replay are demonstrated.
- ☐ Source credentials and network path pass security review.
- ☐ Monitoring, runbook, SLO and support owner are active.
- ☐ Monthly cost range and unit metric are approved.

E.2 Data Product Certification

- ☐ Product owner, steward and consumer are named.
- ☐ Grain, keys, semantics and source authority are documented.
- ☐ Contract, lineage and glossary relationships are complete.
- ☐ Mandatory quality rules pass.
- ☐ Classification, access, masking and encryption controls pass negative tests.
- ☐ Freshness, availability and recovery meet the product tier.
- ☐ Cost allocation and budget are active.
- ☐ Versioning, deprecation and consumer notification are defined.
- ☐ Marketplace or API publication is approved.
- ☐ Quarterly recertification is scheduled.

E.3 Production Readiness

- ☐ Infrastructure and schema deployed from source control.
- ☐ Regional service availability, quotas and limits validated.
- ☐ Failure, duplicate, late, malformed and replay tests passed.
- ☐ Dashboards and alerts route to the correct on-call group.
- ☐ Runbook includes stop, replay, backfill, rollback and escalation.
- ☐ Recovery exercise meets RTO/RPO.
- ☐ Security findings have no unresolved critical/high issues.
- ☐ Data-product scorecard is visible.
- ☐ Consumer acceptance and support handoff are recorded.
- ☐ Post-deployment validation and rollback criteria are automated.

Appendix F. Official Google Cloud Reference Sources

The architecture was aligned to the following official Google Cloud documentation current at publication. Implementation teams must revalidate product status, region support, quotas and limitations at design approval and before production rollout.

Knowledge Catalog overview: docs.cloud.google.com/dataplex/docs/introduction

Knowledge Catalog metadata management: docs.cloud.google.com/dataplex/docs/catalog-overview

Knowledge Catalog data lineage: docs.cloud.google.com/dataplex/docs/about-data-lineage

Knowledge Catalog automatic data quality: docs.cloud.google.com/dataplex/docs/auto-data-quality-overview

Knowledge Catalog data profiling: docs.cloud.google.com/dataplex/docs/data-profiling-overview

BigQuery overview: docs.cloud.google.com/bigquery/docs/introduction

BigQuery partitioned tables: docs.cloud.google.com/bigquery/docs/partitioned-tables

BigQuery clustered tables: docs.cloud.google.com/bigquery/docs/clustered-tables

BigQuery Storage Write API: docs.cloud.google.com/bigquery/docs/write-api

BigQuery row-level security: docs.cloud.google.com/bigquery/docs/row-level-security-intro

BigQuery column-level security: docs.cloud.google.com/bigquery/docs/column-level-security-intro

BigQuery dynamic data masking: docs.cloud.google.com/bigquery/docs/column-data-masking-intro

BigQuery authorized views: docs.cloud.google.com/bigquery/docs/authorized-views

BigQuery sharing: docs.cloud.google.com/bigquery/docs/analytics-hub-introduction

BigLake external tables: docs.cloud.google.com/bigquery/docs/biglake-intro

Apache Iceberg managed tables: docs.cloud.google.com/bigquery/docs/biglake-iceberg-tables-in-bigquery

Lakehouse for Apache Iceberg: docs.cloud.google.com/lakehouse/docs/introduction

Bigtable overview: docs.cloud.google.com/bigtable/docs/overview

Bigtable schema design: docs.cloud.google.com/bigtable/docs/schema-design

Spanner documentation: docs.cloud.google.com/spanner/docs

AlloyDB overview: docs.cloud.google.com/alloydb/docs/overview

Cloud Storage object holds: docs.cloud.google.com/storage/docs/object-holds

Cloud Storage Bucket Lock: docs.cloud.google.com/storage/docs/bucket-lock

Cloud Storage classes: docs.cloud.google.com/storage/docs/storage-classes

Appendix G. Core Google Cloud Service Reference Cards

These reference cards summarize the production role of the principal services used by this volume. They are architecture selection aids, not replacements for service-specific design reviews. Implementation teams must validate current quotas, regional support, preview or general-availability status, VPC Service Controls compatibility, CMEK support and pricing before production approval.

G.1 Cloud Storage

Assessment area	Enterprise guidance
Purpose	Durable object storage for landing, quarantine, immutable raw, replay and archive data.
Architecture role	Provides the evidence layer independent of transformation engines and supports open formats such as Parquet and Avro.
Benefits	Very high durability, lifecycle controls, object versioning, retention policies, event integration and broad ecosystem support.
Limitations	Object storage is not a transactional table engine; small-file proliferation, listing behavior and cross-region transfer require design.
Security	Use uniform bucket-level access, public-access prevention, least-privilege service accounts, VPC Service Controls where supported and CMEK for risk-qualified data.
Monitoring	Monitor request errors, latency, object growth, lifecycle transitions, retention exceptions and unexpected egress.
Cost optimization	Choose region and class from access and residency; compact small files; expire transient landing objects; prevent unnecessary replication.
Alternatives	BigQuery native storage for analytical tables; external object stores for portability; on-premises object storage for retained legacy workloads.
Common pitfalls	Treating folders as security boundaries, sharing object-admin roles, storing millions of tiny files and omitting restore tests.

Production readiness evidence. Approved service configuration, IAM and network design, quota validation, monitoring dashboard, alert routes, recovery procedure, cost owner and service-specific load and security test results.

G.2 BigQuery

Assessment area	Enterprise guidance
Purpose	Primary enterprise analytical warehouse and serving layer for standardized, gold, curated and semantic data products.
Architecture role	Separates managed storage and compute, supports SQL analytics, governance, sharing and integration with BI and AI.
Benefits	Serverless operation, elastic analytical scale, fine-grained security, reservations, native data sharing and integration with Google Cloud data services.
Limitations	Poor partition and query design can create cost and performance problems; location, quota and concurrency constraints require workload planning.
Security	Use dataset boundaries, IAM groups, policy tags, row policies, dynamic masking, authorized interfaces, audit logs and CMEK where required.
Monitoring	Monitor bytes processed, slot use, queue time, query failures, freshness, storage growth, policy changes and cost by product.
Cost optimization	Partition and cluster from observed predicates; materialize repeated expensive logic; separate workloads; enforce budgets and query controls.
Alternatives	Apache Iceberg for multi-engine tables; Bigtable, Spanner or AlloyDB for operational serving; third-party warehouses where mandated.
Common pitfalls	One global dataset, date-sharded tables, recurring SELECT *, direct base-table grants and purchasing reservations without workload evidence.

Production readiness evidence. Approved service configuration, IAM and network design, quota validation, monitoring dashboard, alert routes, recovery procedure, cost owner and service-specific load and security test results.

G.3 BigLake

Assessment area	Enterprise guidance
Purpose	Governed access layer for data in object storage and lakehouse patterns.
Architecture role	Extends BigQuery governance to external data while reducing direct bucket access for consumers.
Benefits	Centralized access control, interoperability with open data and separation of storage ownership from consumer permissions.
Limitations	Performance and feature behavior differ from native tables; metadata, file layout and supported feature combinations must be tested.
Security	Grant access through BigLake connections and governed tables rather than raw bucket permissions; isolate connection identities.
Monitoring	Monitor query latency, bytes scanned, file counts, metadata failures, connection errors and underlying object access.
Cost optimization	Use columnar files, compaction, partition pruning and caching where available; prefer native BigQuery when interoperability is not required.
Alternatives	BigQuery native tables; direct external tables; Apache Iceberg managed tables.
Common pitfalls	Using BigLake for every dataset, permitting consumers to bypass governed tables and ignoring object-layout optimization.

Production readiness evidence. Approved service configuration, IAM and network design, quota validation, monitoring dashboard, alert routes, recovery procedure, cost owner and service-specific load and security test results.

G.4 Apache Iceberg Managed Tables

Assessment area	Enterprise guidance
Purpose	Open table format for workloads that require multi-engine interoperability or customer-controlled object storage.
Architecture role	Provides table metadata, snapshots and open-format access while integrating with BigQuery lakehouse capabilities.
Benefits	Open ecosystem, schema evolution, partition evolution and reduced dependence on a single execution engine.
Limitations	Writer compatibility, catalog authority, compaction, snapshot retention and feature support add operational complexity.
Security	Restrict approved writers, protect catalog and storage identities, enforce data classification and review cross-engine access paths.
Monitoring	Monitor commit failures, metadata growth, snapshot count, small files, orphan files, compaction and engine compatibility.
Cost optimization	Use only for justified interoperability; establish one catalog authority; compact files; expire snapshots according to recovery needs.
Alternatives	BigQuery native tables for managed analytics; unmanaged Iceberg on external engines; duplicate domain-specific copies.
Common pitfalls	Multiple uncontrolled writers, long-lived orphan snapshots, untested engine versions and adopting Iceberg solely as a trend.

Production readiness evidence. Approved service configuration, IAM and network design, quota validation, monitoring dashboard, alert routes, recovery procedure, cost owner and service-specific load and security test results.

G.5 Pub/Sub

Assessment area	Enterprise guidance
Purpose	Managed event backbone for streaming ingestion, fan-out and decoupled producer-consumer integration.

Assessment area	Enterprise guidance
Architecture role	Carries versioned business and technical events to independent ingestion, processing, replay and monitoring consumers.
Benefits	Elastic managed messaging, independent subscriptions, retention, dead-letter handling and integration with Dataflow and BigQuery.
Limitations	Delivery is at least once; ordering and exactly-once subscription features have scope and design implications; payload size and quotas apply.
Security	Use dedicated publisher/subscriber identities, schema controls, topic-level IAM, service perimeters where applicable and encryption controls.
Monitoring	Monitor oldest unacked age, backlog, delivery attempts, publish errors, throughput, DLQ volume and subscription health.
Cost optimization	Right-size retention, filter subscriptions, avoid unnecessary fan-out and compress or externalize oversized payloads.
Alternatives	Managed Kafka for Kafka-native requirements; direct API calls for synchronous interaction; batch files for non-event workloads.
Common pitfalls	Assuming exactly-once business processing without idempotency, shared subscriptions, unbounded retention and no replay ownership.

Production readiness evidence. Approved service configuration, IAM and network design, quota validation, monitoring dashboard, alert routes, recovery procedure, cost owner and service-specific load and security test results.

G.6 Dataflow

Assessment area	Enterprise guidance
Purpose	Managed Apache Beam execution for complex batch and streaming processing.
Architecture role	Implements stateful processing, windowing, enrichment, deduplication, validation and scalable file transformation.
Benefits	Unified programming model, autoscaling, managed worker lifecycle, templates and broad connector ecosystem.
Limitations	Pipeline engineering and state semantics are more complex than SQL; worker networking, quotas and hot keys require design.
Security	Use private workers, least-privilege service accounts, encrypted staging and temporary locations, restricted images and approved dependencies.
Monitoring	Monitor system lag, data freshness, throughput, worker CPU and memory, autoscaling, failed bundles, hot keys and DLQ rates.
Cost optimization	Use autoscaling, efficient coders, right-size workers, avoid unnecessary shuffles and select streaming only when justified.
Alternatives	BigQuery SQL or Dataform for SQL transformations; Serverless Spark for Spark workloads; Cloud Run for simple stateless processing.
Common pitfalls	Embedding business rules without contracts, ignoring event time, non-idempotent sinks, hard-coded secrets and no replay testing.

Production readiness evidence. Approved service configuration, IAM and network design, quota validation, monitoring dashboard, alert routes, recovery procedure, cost owner and service-specific load and security test results.

G.7 Datastream

Assessment area	Enterprise guidance
Purpose	Serverless change-data-capture service for supported operational databases and targets.
Architecture role	Captures inserts, updates and deletes with source metadata for low-latency ingestion into approved landing or analytical patterns.
Benefits	Reduces connector operations, supports continuous replication and integrates with Google Cloud data services.
Limitations	Source and target coverage, DDL behavior, backfill, network connectivity

Assessment area	Enterprise guidance
	and schema evolution must be validated per workload.
Security	Use private connectivity where supported, dedicated database accounts, least-privilege destination identities and encrypted secrets.
Monitoring	Monitor stream state, lag, errors, rejected records, source log retention, backfill progress and reconciliation variance.
Cost optimization	Filter unnecessary objects and columns, tune backfill windows and avoid replicating data without consumer demand.
Alternatives	Debezium and Kafka for custom or unsupported CDC; third-party CDC; scheduled incremental extracts.
Common pitfalls	Treating CDC as source truth without reconciliation, allowing source logs to expire, ignoring deletes and deploying without schema-change tests.

Production readiness evidence. Approved service configuration, IAM and network design, quota validation, monitoring dashboard, alert routes, recovery procedure, cost owner and service-specific load and security test results.

G.8 Storage Transfer Service

Assessment area	Enterprise guidance
Purpose	Managed bulk and recurring transfer of objects from on-premises, Google Cloud and supported cross-cloud sources.
Architecture role	Moves historical and scheduled object data into controlled landing or raw zones with transfer evidence.
Benefits	Managed scheduling, incremental transfer, checksums, agent-based options and cross-cloud support.
Limitations	Network bandwidth, source API limits, object metadata differences and egress charges can dominate design.
Security	Use dedicated agents and identities, least privilege, private connectivity where available, encrypted transport and transfer logs.
Monitoring	Monitor transfer errors, skipped objects, checksum mismatch, throughput, backlog, source throttling and egress.
Cost optimization	Pre-size bandwidth, avoid duplicate copies, schedule around source limits and apply lifecycle after reconciliation.
Alternatives	Transfer Appliance for very large offline transfer; custom copy tools; partner integration services.
Common pitfalls	Migrating without inventory, no restart or reconciliation plan, underestimating egress and treating transfer completion as data validation.

Production readiness evidence. Approved service configuration, IAM and network design, quota validation, monitoring dashboard, alert routes, recovery procedure, cost owner and service-specific load and security test results.

G.9 Cloud Run

Assessment area	Enterprise guidance
Purpose	Serverless container platform for API ingestion, lightweight connectors, validation services and data-product APIs.
Architecture role	Provides stateless integration endpoints where managed connectors or data-processing engines are not the right fit.
Benefits	Rapid deployment, autoscaling, container portability, identity integration and request or event-driven execution.
Limitations	Request duration, concurrency, startup, outbound connectivity and state management require design; not a replacement for long-running data engines.
Security	Require authenticated invocation, dedicated service accounts, Secret Manager, controlled egress and private ingress where appropriate.
Monitoring	Monitor latency, error rate, instance count, concurrency, cold starts,

Assessment area	Enterprise guidance
	outbound failures and dependency SLOs.
Cost optimization	Set concurrency and minimum instances from latency need; avoid polling; scale to zero for intermittent workloads.
Alternatives	Cloud Functions for small event handlers; Workflows for orchestration; GKE for specialized runtime or control needs.
Common pitfalls	Embedding credentials, storing state locally, unbounded retries, public endpoints by default and using Cloud Run for massive data shuffles.

Production readiness evidence. Approved service configuration, IAM and network design, quota validation, monitoring dashboard, alert routes, recovery procedure, cost owner and service-specific load and security test results.

G.10 Knowledge Catalog

Assessment area	Enterprise guidance
Purpose	Enterprise metadata, discovery and governance surface for Google Cloud data assets.
Architecture role	Publishes technical and business metadata, glossary terms, lineage, profiling, quality results, ownership and product certification.
Benefits	Native asset inventory, search, governance metadata and integration with BigQuery and Cloud Storage.
Limitations	Coverage varies by asset and lineage source; metadata quality still depends on ownership and operating process.
Security	Restrict metadata administration, protect sensitive descriptions, separate catalog roles and audit governance changes.
Monitoring	Monitor metadata completeness, stale ownership, lineage coverage, failed scans, quality execution and certification age.
Cost optimization	Automate metadata from contracts and deployments; avoid manual-only cataloging; tier scans by product criticality.
Alternatives	Third-party enterprise catalog integrated with Google Cloud; custom metadata repository; spreadsheets only for temporary discovery.
Common pitfalls	Cataloging everything without ownership, exposing sensitive metadata, treating glossary terms as optional and certifying without evidence.

Production readiness evidence. Approved service configuration, IAM and network design, quota validation, monitoring dashboard, alert routes, recovery procedure, cost owner and service-specific load and security test results.

G.11 BigQuery Sharing

Assessment area	Enterprise guidance
Purpose	Governed publication and subscription mechanism for analytical data and supported streams.
Architecture role	Implements an enterprise data marketplace without unmanaged extracts or broad producer-project access.
Benefits	Publisher control, linked datasets, cross-project or cross-organization discovery and reduced data copying.
Limitations	Location, object-type and feature restrictions apply; subscriber governance and product contracts remain necessary.
Security	Publish approved interfaces only, control exchange and listing administration, monitor subscribers and retain producer revocation capability.
Monitoring	Monitor listing changes, subscriptions, query usage, failed access, product freshness and unexpected consumer patterns.
Cost optimization	Prefer linked access to copies; publish only demanded products; retire unused listings and attribute producer costs.
Alternatives	Authorized views or datasets for internal sharing; exports for contractual

Assessment area	Enterprise guidance
Common pitfalls	offline exchange; APIs for operational interfaces.
	Publishing raw datasets, omitting consumer terms, ignoring location alignment and assuming sharing transfers data-product support responsibility.

Production readiness evidence. Approved service configuration, IAM and network design, quota validation, monitoring dashboard, alert routes, recovery procedure, cost owner and service-specific load and security test results.

G.12 Bigtable

Assessment area	Enterprise guidance
Purpose	Operational wide-column store for high-throughput, low-latency key-based access.
Architecture role	Serves selected operational data products such as telemetry, time series, personalization or feature access.
Benefits	Horizontal scale, low-latency access, replication options and strong fit for sparse key-value or time-series patterns.
Limitations	No general relational joins; row-key design can create hotspots; query flexibility is intentionally limited.
Security	Use application profiles, IAM, private access, CMEK where required and tenant-aware row-key or access patterns.
Monitoring	Monitor node or cluster CPU, storage, latency, hotspot indicators, replication, garbage collection and request errors.
Cost optimization	Design row keys from access patterns, use autoscaling where appropriate and apply garbage-collection policies.
Alternatives	Spanner for relational transactions; BigQuery for analytics; cache services for ephemeral data.
Common pitfalls	Sequential hot keys, scanning without key bounds, treating Bigtable as a warehouse and failing to load-test the key design.

Production readiness evidence. Approved service configuration, IAM and network design, quota validation, monitoring dashboard, alert routes, recovery procedure, cost owner and service-specific load and security test results.

G.13 Spanner

Assessment area	Enterprise guidance
Purpose	Globally scalable relational database for mission-critical transactional applications needing strong consistency.
Architecture role	Supports operational data products that require relational semantics, high availability and scale beyond conventional regional databases.
Benefits	Managed distributed transactions, relational SQL, horizontal scale and multi-region configurations.
Limitations	Schema and query design differ from traditional databases; configuration and cost must match real global requirements.
Security	Use database roles and IAM, private access, CMEK where required, least-privilege applications and audit logs.
Monitoring	Monitor CPU, latency, storage, lock contention, query statistics, split behavior, replication and availability.
Cost optimization	Select configuration deliberately, use interleaving and indexes carefully, tune queries and avoid overprovisioning.
Alternatives	AlloyDB or Cloud SQL for regional PostgreSQL; Bigtable for key-value; BigQuery for analytics.
Common pitfalls	Selecting Spanner for brand value, importing monolithic schemas unchanged and ignoring commit-latency geography.

Production readiness evidence. Approved service configuration, IAM and network design, quota validation, monitoring dashboard, alert routes, recovery procedure, cost owner and service-specific load and security test results.

G.14 AlloyDB

Assessment area	Enterprise guidance
Purpose	Managed PostgreSQL-compatible database for demanding regional operational workloads.
Architecture role	Hosts operational applications, serving APIs and metadata services that need PostgreSQL compatibility and strong performance or HA.
Benefits	PostgreSQL ecosystem compatibility, managed high availability, read pools and analytical acceleration capabilities.
Limitations	Regional database characteristics remain; connection management, failover and cross-region DR require design.
Security	Use private IP, database IAM where suitable, Secret Manager, CMEK where required and least-privilege database roles.
Monitoring	Monitor CPU, memory, storage, connections, replication lag, query latency, failover and backup status.
Cost optimization	Use connection pooling, right-size read pools, tune indexes and queries and manage backup retention.
Alternatives	Cloud SQL for simpler workloads; Spanner for global consistency or scale; Bigtable for key-value serving.
Common pitfalls	Direct internet exposure, unlimited connections, using replicas as backup and assuming PostgreSQL compatibility eliminates performance testing.

Production readiness evidence. Approved service configuration, IAM and network design, quota validation, monitoring dashboard, alert routes, recovery procedure, cost owner and service-specific load and security test results.

G.15 Cloud KMS and Cloud HSM

Assessment area	Enterprise guidance
Purpose	Central key-management services for CMEK and cryptographic-control requirements.
Architecture role	Provide separation of duties and customer-controlled key lifecycle for restricted data-platform assets.
Benefits	Central policy, auditability, rotation, hardware-backed keys through Cloud HSM and integration with supported services.
Limitations	Key unavailability can stop data access; location, service integration, permissions and rotation procedures require coordination.
Security	Separate key administrators from service users, restrict encrypt or decrypt use, monitor key access and protect deletion schedules.
Monitoring	Monitor denied access, unusual encrypt or decrypt use, disabled versions, rotation age and dependent-service failures.
Cost optimization	Use risk-based CMEK, minimize cross-project complexity and avoid keys where no control requirement exists.
Alternatives	Google-managed encryption for standard workloads; External Key Manager for external control requirements.
Common pitfalls	Universal CMEK without operating ownership, deleting keys before data expiry and granting broad cryptographic roles.

Production readiness evidence. Approved service configuration, IAM and network design, quota validation, monitoring dashboard, alert routes, recovery procedure, cost owner and service-specific load and security test results.

G.16 Sensitive Data Protection

Assessment area	Enterprise guidance
Purpose	Discovery, inspection and transformation of sensitive data such as personal identifiers and regulated content.
Architecture role	Supports classification, tokenization, de-identification and privacy controls in ingestion and governance workflows.

Assessment area	Enterprise guidance
Benefits	Managed detectors, custom info types, scalable inspection and transformation patterns.
Limitations	False positives and negatives require tuning; inspection cost and data movement must be controlled; it does not replace access governance.
Security	Use scoped service identities, protected templates, regional processing validation and strict access to findings.
Monitoring	Monitor job failures, inspection coverage, finding trends, template changes, transformation errors and cost.
Cost optimization	Scan representative samples and high-risk assets first; reuse templates; avoid rescanning unchanged large data without purpose.
Alternatives	Application-native validation, custom classifiers and third-party privacy tools.
Common pitfalls	Treating detection as proof of absence, storing findings with broad access and tokenizing without re-identification-key governance.

Production readiness evidence. Approved service configuration, IAM and network design, quota validation, monitoring dashboard, alert routes, recovery procedure, cost owner and service-specific load and security test results.

Closing Recommendation

Implement the platform in capability slices rather than launching all services simultaneously. Establish the source registry, contract model, layer standards, catalog, access model, quality gates, product scorecard and cost allocation before scaling ingestion. Deliver one priority domain that exercises batch, CDC and streaming only where justified. Certify a small number of products, prove replay and recovery, measure unit cost, then expand through reusable modules and domain ownership.

The platform is successful when consumers choose governed products because they are easier and more reliable than bypassing the platform. Architecture controls must therefore be automated, observable and proportional to risk. Governance that cannot be executed in engineering workflows will be ignored; engineering that cannot provide evidence will not be trusted.